Project title	Pantanal REDD+ Program
Project ID	4747
Monitoring period	14-October-2021 to 31-December-2023
Crediting period	14-October-2021 to 13-October-2061
Original date of issue	19-December-2024
Most recent date of issue	12-January-2024
Version	1.3 draft version
VCS Standard Version	VCS Standard, v4.5
Prepared by	Biofílica Ambipar Environment Investments

CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

The Pantanal biome is located in the center-west of Brazil, covering portions of the states of Mato Grosso and Mato Grosso do Sul. Considered the largest continuous freshwater wetland on the planet, this biome oscillates between terrestrial and aquatic phases, forming a mosaic of landscapes that provide great biodiversity of flora and fauna. Analyzing the historical context of the region where the Pantanal is located, there has been a considerable expansion of agricultural activities (the main local economic activity, mainly focused on the breeding of Pantanal cattle) and this factor, together with the high rates of forest fires in this area, results in a scenario of unsustainable exploitation of the natural resources of this biome.

Considering all the applicable legal requirements, landowners who have a surplus of native vegetation on their property and choose to preserve it, would fulfill the first requirement for obtaining economic incentives from the voluntary carbon market, through Reducing Emissions from Deforestation and Forest Degradation (REDD) through Avoided Planned Deforestation (APD). Given all this context, the implementation of the REDD+ Carbon Program on a scale for the Pantanal biome is possible, providing a model of sustainable economic development that can help to redress the balance, reconciling the conservation of environmental services with traditional economic activities. Biofílica Ambipar, in partnership with rural landowners in the Pantanal of Mato Grosso do Sul who are committed to promoting dialogue that benefits both the environment and sustainable economic development, aims to develop economic activities linked to the conservation of environmental services and all their ecological functions through the implementation of the Pantanal REDD+ Program.

The first instance of the REDD+ Pantanal Program, made up of 11 private properties located in the municipalities of Aquidauana, Corumbá and Miranda, in the state of Mato Grosso Sul, has as its starting date the purchase of the Santa Sofia farm, which took place on October 14, 2021, an event that marks the beginning of conservation actions in the region where the project is located.

The REDD+ Pantanal Program has a credit period of 50 years (40 years per instance), with a useful life and longevity of 50 years in which the project's activities will be maintained, including: conducting socio-economic, biodiversity and carbon stock diagnoses, mitigating leakage through continuous monitoring, holding continuous training workshops for workers on REDD+ and other topics of interest, monitoring biodiversity and endangered species, monitoring deforestation and emissions through satellite images, property surveillance, among others.

The implementation of the first instance of the Pantanal REDD+ Program will prevent the deforestation of 23,269 hectares of phytophysiognomies of forests and forested savannas and, in its first reference period (10 years), will reduce emissions by 1,887,177 tCO2e, representing an average annual reduction in GHG emissions of 187,718 tons of CO_2.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	14-October-2021 to 13-October-2071	VCS & CCB	In the process of being hired	50 years
Verification	14-October-2021 to 31-December-2023	VCS	In the process of being hired	2 years

1.3 Sectoral Scope and Project Type

Sectoral scope	14 - Agriculture, Forestry and Other Land Uses (AFOLU)
AFOLU project category	Reducing Emissions from Deforestation and Forest Degradation (REDD)
Project activity type	Avoided Planned Deforestation (APD)

1.4 Project Eligibility

1.4.1 General eligibility

The REDD+ Pantanal Program has the following eligibility criteria:

1) The project complies with all applicable rules and requirements set out in the VCS Program;

2) The project follows parameters established in an eligible methodology under the VCS Program, VM0007 (more details in section 3. Application of the Methodology);

3) The implementation of the activities of this project does not lead to the violation of any law applicable to its region of insertion (more details in section 1.15 Compliance with laws, statutes and other regulatory frameworks);

4) It is not located in a jurisdiction covered by a jurisdictional REDD+ program.

1.4.2 AFOLU project eligibility

This is an eligible project in the AFOLU (Agriculture, Forestry and Other Land Use) category under the VCS Program through the reduction of emissions from deforestation and degradation (REDD) within the framework of avoided planned deforestation (APD).

According to data collected from the MapBiomas platform, collection 8[1] the Pantanal REDD+ Program area is eligible as forest or savannah that has been forested for more than 10 years and, therefore, there has been no conversion of the ecosystem in order to generate carbon credits.

1.4.3 Transfer project eligibility

Not applicable.

1.5 Project Design

Indicate if the project has been designed as:

☐ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☒ Grouped project

Grouped Project Design

This Program was designed as a grouped project, under a set of eligibility criteria specified in the VCS Standard v4.5, relating to 3 major topics described below, these being: baseline scenario and additionality; new activity instances; and inclusion of new activity instances in the project.

As this is a grouped project, the expansion of its scope is allowed through the inclusion of new instances after the initial validation of the project, occurring at each verification event, as long as all the eligibility criteria are duly met.

1.5.1 Baseline Scenario and Additionality

i. *Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed. Such geographic areas shall be specified using geodetic polygons as set out in Section 3.11 below. Geographic areas with no initial project activity instances shall not be included in the project unless it can be demonstrated that the same (or at least as conservative) baseline scenario and rationale for the demonstration of additionality is applicable to such an area as a geographic area that does include initial project activity instances:*

   To delimit the geographical area of the Pantanal REDD+ Program, the Brazilian Pantanal biome as a whole was considered, excluding from this limit legally protected areas where legal deforestation would not be able to occur, these being: Conservation Units, Indigenous Lands and

[1] https://brasil.mapbiomas.org/

Quilombola Territories. In this way, the geographical area of this Program extends across the states of Mato Grosso and Mato Grosso do Sul in Brazil, covering around 16 municipalities where the REDD+ Pantanal Program can operate.

Eligible to become instances of the Pantanal REDD+ Program are legally established private properties with surplus vegetation classified as forest and/or forested savannah according to the Mapbiomas Collection 8 classification, and which can be legally deforested according to the environmental legislation in force at the national, state and municipal levels.

To meet the minimum vegetation cover required by law, properties must conserve a percentage of 20% of Legal Reserve and, for properties that fall within the Restricted Use Area of the Pantanal Floodplain, 50% of vegetation cover classified as "Cerrado" formations with a high density of trees (Forested Savannah and Wooded Savannah) and 40% of vegetation cover classified as grassland formations (Park Savannah, Woody Grass Savannah and Steppic-Savannah) must be preserved.

Further details on the geographical area can be found in section 1.13 Project location, and a KML file containing the geodetic polygon clearly defining the boundaries of the geographical area into which this Program can expand, as well as the boundaries of the project area, will be provided together with its documentation.

ii. *Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description. The initial project activity instances may also include any instances of the project activity that have been planned and developed to a sufficient level of detail to enable their assessment at validation:*

In order to determine the baseline scenario and demonstrate the additionality of the Pantanal REDD+ Program, it was considered that all the properties within its geographical area are subject to the same legal framework (with peculiarities regarding legislation at state level) and have the same common practice scenario that dominates the Pantanal biome, which would be the conversion of land use from forested areas to livestock, mainly aimed at raising cattle.

More details on the baseline and additionality scenarios can be found in sections 3.4 and 3.5, respectively.

iii. *As with non-grouped projects, grouped projects may incorporate multiple project activities (see Section 3.6.1 – 3.6.3 for more information on multiple project activities). Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area:*

The Pantanal REDD+ Program has only one project activity, which is restricted to avoided planned deforestation (APD), following the guidelines stipulated by the methodology VM0007 v1.6.

In order to guarantee the conservation of the excess vegetation on the properties participating in this Program, a long-term contract has been signed between Biofílica Ambipar Environments and each of the landowners. In addition, social and biodiversity activities will be implemented, as well as activities aimed at mitigating leaks and monitoring activities, which are described in sections 1.12 and 1.18.1.

iv. *The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project. Where a single baseline scenario cannot be determined for a project activity over the entirety of a geographic area, the geographic area shall be redefined or divided such that a single baseline scenario can be determined for the revised geographic area or areas:*

The Pantanal REDD+ Program defines only one baseline that is applied to the entire extent of its geographical area and follows all the criteria stipulated by the VM0007 v1.6 methodology, as specified in item 3.4 Baseline Scenario.

v. *The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the methodology applied to the project. Where the additionality of the initial project activity instances within a particular geographic area cannot be demonstrated for the entirety of that geographic area, the geographic area shall be redefined or divided such that the additionality of the instances occurring in the revised geographic area or areas can be demonstrated:*

The Pantanal REDD+ Program has only one defined geographical area and its additionality is shown for its total extension, following all the criteria stipulated by the VM0007 v1.6 methodology. More details can be found in section 3.5 Additionality.

vi. *Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rates:*

All the relevant factors for determining the baseline and demonstrating the additionality of the Pantanal REDD+ Program were scaled for the entire geographical area of this Program.

1.5.2 New Project Activity Instance Eligibility Criteria

For instances to be included in a grouped project, it must meet one or more of the eligibility criteria listed below:

i. *Meet the applicability conditions set out in the methodology applied to the project:*

As presented in section 3.2 Applicability of the methodology, the Pantanal REDD+ Program follows all the requirements of the VM0007 v1.6 methodology, an eligible methodology under the VCS Program.

With regard to the instances included in this Program, all participating properties meet the following conditions of applicability:

- The project insertion area is not registered with the Clean Development Mechanism (CDM) or any other Greenhouse Gas (GHG) program;
- The implementation area of the project is classified as forest vegetation according to the Brazilian forest definition, and also includes areas classified as Savannah Forest, which are areas with a significant density of trees, but without the formation of a canopy. The main basis for delimiting these phytophysiognomies in the project area was the MapBiomas platform, collection 8;
- The baseline scenario for inserting the properties falls under the concept of avoided planned deforestation (APD);
- The baseline scenario is based on the legal conversion of land use from forested to non-forested areas, according to the environmental legislation in force at national, state and municipal level in the project area (Brazil, states of Mato Grosso and Mato Grosso do Sul);
- Leak prevention activities do not include: flooded agricultural land (such as areas for growing rice, for example); and areas where there is intensive livestock production through confinement, controlled diets to achieve rapid weight gain and/or the use of manure lagoons (a place where liquid manure produced by animals is stored).

ii. *Use the technologies or measures specified in the project description:*

The reduction in GHG emissions will be the effect of signing a long-term forest conservation agreement with the owners of the properties that are part of the current and future instances of this project. Complementary activities related to biodiversity conservation, social activities, as well as monitoring activities have been described in section 1.18 Contributions to sustainable development and in section 6 Monitoring.

iii. *Apply the technologies or measures in the same manner as specified in the project description:*

Current and future instances of this project must apply the same technologies and measures specified in this document, and any adjustments/changes will be reported and duly described.

iv. *Are subject to the baseline scenario determined in the project description for the specified project activity and geographic área:*

The only geographical area of the Pantanal REDD+ Program is the Pantanal biome, which partially covers the states of Mato Grosso and Mato Grosso do Sul, including 16 municipalities. Item 1.13 Location of the project provides more details on this subject.

New instances must include legally constituted private properties that have forest cover that can be legally deforested, meeting the criteria for avoided planned deforestation (APD).

v. *Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances:*

The new instances that will be part of this project should follow the same additionality approach presented in section 3.5.1 Identification of alternative land use scenarios.

1.5.3 Inclusion of New Project Activity Instances

Grouped projects provide for the inclusion of new project activity instances subsequent to the initial validation of the project. New project activity instances shall:

- *Occur within one of the designated geographic areas specified in the project description:*

  New instances to be added will be restricted to within the previously established boundary of the only geographical area of the Pantanal REDD+ Program, which is based on the boundary of the Pantanal biome.

- *Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient:*

  New instances that are added to the Pantanal REDD+ Program will meet the following eligibility criteria:

  - Rural properties within the boundaries of the Pantanal biome, more specifically within the geographical area stipulated for the Pantanal REDD+ Program;
  - Commitment to the long-term conservation of the surplus forest cover on the property, agreed through the signing of a long-term contract between Biófilica Ambipar Environment and the owner of the land where the project will be implemented;
  - Rural Environmental Registry of the property with "Regular", "Active" or "Registered for analysis" status in the National Rural Environmental Registry System (SICAR) and the IMASUL System for Registration and Strategic Information on the Environment (SIRIEMA), the latter only if the rural property in question is located in the state of Mato Grosso do Sul.

- *Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body:*

The monitoring report for new instances will be based on the VCS Monitoring Report v4.3 template or the most up-to-date version available at the time, covering all the eligibility criteria stipulated in this item and providing evidence of all the activities planned for the Pantanal REDD+ Program.

- *Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions):*

  The proponent of the Pantanal REDD+ Program (Biofílica Ambipar Environment) has a partnership agreement with the owners of the land where the project is being implemented. Since it is the landowners who have the right to use and possess the farms and their resources, a long-term contract has been signed between the parties, aimed at forest conservation in the areas targeted by this Program.

  The current owners of the farms included in the first instance of this Program have remained the same since the start date stipulated for the project, October 14, 2021. All documents proving ownership of the project areas are kept under the management of the proponent and will be duly presented to the VVB during the validation/verification process.

- *Have a start date that is the same as or later than the grouped project start date:*

  Future instances to be incorporated into the Pantanal REDD+ Program will have their criteria for definition of the start date duly justified, following this stipulated eligibility criterion.

- *Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period:*

  The crediting period for new instances will be defined according to the baseline scenario stipulated for the project, which is based on the project area and the planned deforestation rate. Further details on the baseline can be found in section 3.4 Baseline Scenario.

- *Not be or have been enrolled in another VCS project:*

  The Pantanal REDD+ Program has not been registered with any other GHG emissions program, nor have any of its current or future instances.

- *Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9 of VCS Standard v4.5:*

  Transcribing items 3.6.8 and 3.6.9 of the VCS Standard v4.5, we have:

  - *3.6.8 The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other):*

All instances of activity within a 10 km radius will be included in a single project, covered by the Pantanal REDD+ Program.

- *3.6.9 Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows:*

  1) Each project activity instance that exceeds one percent of the capacity limit shall be identified.

  2) Such instances shall be divided into clusters, whereby each cluster is comprised any system of such instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.

  3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.

  The capacity limit does not apply to the Pantanal REDD+ Program.

- *When the inclusion of a new project activity instance includes the addition of a new project proponent, such instances must be included in the description of the grouped project within five years (in the case of AFOLU projects) of the start date of the project activity instance:*

  The REDD+ Pantanal Program has Biofílica Ambipar Environment Investments as its sole proponent and it is not intended to include another proponent in this Program.

1.6 Project Proponent

Organization name	Biofílica Ambipar Environment Investments
Contact person	Plínio Ribeiro
Title	Chief Executive Officer (CEO)
Address	Rua Vieira de Moraes, 420 – conjunto 43/44 – São Paulo (SP), Brasil
Telephone	+55 11 3073-0430
Email	plinio@biofilica.com.br

1.7 Other Entities Involved in the Project

Organization name	Instituto Taquari Vivo - ITV

Role in the project	Providing support in the strategic articulation in the region with public-private institutions of interest and supporting the construction of project documentation.
Contact person	Renato Roscoe
Title	Executive Director
Address	Avenida Afonso Pena, 5723 - Sala 1504, Bairro Santa Fé – Campo Grande/MS
Telephone	-
Email	contato@taquarivivo.org

Organization name	Restaura Consultoria Ambiental e Treinamentos LTDA
Role in the project	Preparation of socio-economic and biodiversity diagnoses.
Contact person	Letícia Koutchin Reis
Title	Executive Director
Address	Rua Luis Gama, 121, Bairro Amambai – Campo Grande/MS
Telephone	+55 67 999175884
Email	contato@restauraconsultoria.com

Organization name	STA Soluções Técnicas e Serviços de Engenharia Ambiental LTDA
Role in the project	Preparation of Carbon stock diagnosis.
Contact person	Raniery Vale Neri Branco
Title	Project Development Engineer
Address	Rua vinte e oito de setembro, 1226, Bairro Reduto – Belém/PA
Telephone	-
Email	raniery@ambientalsta.com.br

Organization name	Monteiro de Barros Sociedade de Advogados

Role in the project	Consultancy on legal and environmental issues for the Pantanal REDD+ Program.
Contact person	Theotonio Mauricio Monteiro de Barros
Title	Lawyer
Address	Rua Ferreira de Araújo, 221 – cj. 34, Bairro Alto de Pinheiros – São Paulo/SP
Telephone	+55 11 3039-1819
Email	theotonio@mdbadv.com.br

Organization name	Associação Aliança 5P
Role in the project	Supports the development of the project's activities, representing the properties that are members of the 5P Alliance
Contact person	Ana Paulo Felício
Title	Executive Secretary
Address	Avenida Afonso Pena, 5723 – Sala 1504, Bairro Santa Fé – Campo Grande/MS
Telephone	-
Email	alianca5P@outlook.com

1.8 Ownership

The Pantanal REDD+ Program area is located within the boundaries of private properties that have the right to use and possess the farms and their resources, since they are private properties and have all the documentation that proves their ownership. Following the requirement of item 3.7.1 of VCS Standard 4.5 *"1) Project ownership arising or granted under statute, regulation, or decree by a competent authority"*. There are no disputes over titles or rights in the project area.

The documents proving ownership (land data for each property) will be shared with the VBB on a confidential basis. A list of the documentation submitted can be found in the Appendix: List of land documents collected.

1.9 Project Start Date

Project start date	14-October-2021
Justification	In development. Previous content below: Started by a group of businessmen who made a common commitment to conserve the forest cover on their properties, the Pantanal Conservation Initiative was born, the decisive step being the acquisition of Santa Sofia Farm. This area covers around 34,000 hectares and was at risk of losing 50% of its vegetation cover due to changes in state legislation. The aim of buying Santa Sofia Farm was to consolidate a biodiversity corridor in the Pantanal region, linking the regions ("Pantanais") of Miranda, Aquidauana and Nhecolândia. The Santa Sofia farm was bought by eight shareholders and its registration was transcribed on 10/14/2021 in the name of the Onçafari Association, the institution that is currently responsible for the management and maintenance of this farm. The coming together of people with a mutual interest in the conservation of the Pantanal biome to buy the Santa Sofia farm was the first milestone that gave rise to an association that today is called the 5P Alliance (for Pantanal, Preservation, Partnerships, Livestock and Productivity), a civil association of a private nature, with no economic or non-profit purpose, whose aim is to conduct, support and coordinate studies, projects and initiatives associated with economic sustainability, nature preservation and socio-economic-environmental development in the Pantanal region and neighboring areas. Given this context, it is understood that the effective conservation actions in the regions of the REDD+ Pantanal Program began with the purchase of the Santa Sofia farm, culminating in the organization of farmers with a common commitment to prevent planned deforestation in the Pantanal region.

1.10 Project Crediting Period

Crediting period	Ten years, fixed

Start and end date of first or fixed crediting period	14-10-2021 to 13-October-2031, resulting in a period of 10 years fixed.

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
14-October-2021 to 13-October-2022	168.861
14-October-2022 to 13-October-2023	173.274
14-October-2023 to 13-October-2024	177.686
14-October-2024 to 13-October-2025	182.099
14-October-2025 to 13-October-2026	186.511
14-October-2026 to 13-October-2027	190.924
14-October-2027 to 13-October-2028	195.337
14-October-2028 to 13-October-2029	199.749
14-October-2029 to 13-October-2030	204.162
14-October-2030 to 13-October-2031	208.574
Total estimated ERRs during the first or fixed crediting period	**1.887.177**
Total number of years	**10**
Average annual ERRs	**187.718**

1.12 Description of the Project Activity

Under development. We are awaiting the results of the socio-environmental diagnostics to improve this structure. Previous content below.

Weather	Community	Biodiversity	CLASS	ACTIVITY	DESCRIPTION
			Management	Prospect new areas for inclusion in the Pantanal REDD+ Program	It occurs through 3 main mechanisms: 1) Partnerships with institutions that bring together various rural landowners, such as unions and sector associations; 2) Partnerships with prospectors and area brokers; and 3) Word of mouth (buzz marketing).
			Operational	Conduct stock, socio-economic and environmental diagnoses (DSEA)	Hiring suppliers to carry out technical studies, guidance and monitoring of the activities to be carried out.
			Operational	Carry out study to determine the deforestation baseline	Analysis and understanding of VM0007 methodology guidelines and planned deforestation dynamics for designing the Pantanal REDD+ Program baseline and modeling future deforestation.
			Operational	Hold workshops with researchers and proponents to present results and design activities	Two workshops were held: I Workshop: Preliminary presentation of the diagnostic results; II Workshop: Final presentation of the diagnostic results
			Governance	Establish institutional articulation/ partnerships with public bodies and private entities	Establish institutional partnerships to support the maintenance, expansion and implementation of the project activities.
			Operational	Monitoring deforestation and emissions using satellite images and generating annual reports	Monitoring the dynamics of deforestation using satellite images, which will result in monitoring bulletins that will be shared with interested parties.

Weather	Community	Biodiversity	CLASS	ACTIVITY	DESCRIPTION
			Operational	Carry out surveillance of properties	Inspection activities (patrols, remote monitoring) in the area where the REDD+ Pantanal Program is being implemented.
			Operational	Implement a plan to prevent and combat forest fires on properties	Training with the help of local employees and setting up fire brigades.
			Governance	Carry out leakage mitigation	Monitoring of other properties (not in the project area) that the owners of areas participating in the Pantanal REDD+ Program already own or may acquire in the future, with the main aim of preventing planned deforestation in these areas.
			Operational	Hold ongoing training workshops for workers on REDD+ and topics of interest	Training for all employees of farms and outsourced companies on REDD+ activities, labor rights, the importance of environmental conservation, among other relevant topics.
			Operational	Monitoring biodiversity (fauna and flora) combined with encouraging and enabling scientific research by local institutions	Drawing up a long-term monitoring plan, preferably through agreements with local teaching and research institutions, taking into account the project's impact on regional biodiversity.
			Operational	Preserving and monitoring endangered species	Monitoring through systematic field campaigns.
			Operational	Preserving and monitoring the High Conservation Value Area (HCVA)	Monitoring through systematic field campaigns.
			Management	Carry out permanent quality control	Monitoring the implementation, effectiveness and efficiency of social and environmental management actions.

1.13 Project Location

The region where the REDD+ Program is being implemented is the Pantanal biome, which is located in central-western Brazil, encompassing the states of Mato Grosso and Mato Grosso do Sul (Figure 1). This biome is subdivided into 11 regions: Porto Murtinho, Nabileque, Miranda, Aquidauana, Abobral, Nhecolândia, Paraguay, Paiaguas, Barão de Melgaço, Paconé and Cáceres (Figure 2). It is worth mentioning that the expansion area of the Pantanal REDD+ Program is the entire Pantanal biome, including all of its 11 sub-regions.

The first instance of the REDD+ Pantanal Program is located in Brazil, in the state of Mato Grosso do Sul, more specifically in the municipalities of Aquidauana, Miranda and Corumbá, and falls within the Pantanal sub-regions of Aquidauana, Miranda, Nhecolândia and Abobral, comprising 11 rural properties (with a total area of 138,453 hectares): Barranco Alto, Estância Caiman, Embiara, Fazendinha, Porto Cyríaco, Santa Sofia, Vera Lúcia, Primavera, Eldorado do Rio Negro, Havaí and Salinas. Within these boundaries is the project area of 23,269 hectares. Its surroundings are characterized by the presence of the Pantanal do Rio Negro State Park and the Cachoeirinha Indigenous Land, located near the Rio Vermelho and Rio Negro, as illustrated in Figure 4.

Table 1 details the reference coordinates (center point) of each of the 11 properties inserted in the first instance of the project, as well as listing the numbers indicated on each property that appear in Figure 3. The coordinates of the vertices of each property will be provided via a KML file.

Table 1. Reference coordinates of the properties participating in the Pantanal REDD+ Program

Identification	Property	UTM coordinates	
		X	Y
1	Salinas	594175.4	7876868.5
2	Porto Cyríaco	577479.0	7823764.5
3	Embiara	598474.4	7839314.5
4	Havaí	594330.2	7879131.7
5	Fazendinha	553642.1	7840544.7
6	Barranco Alto	589882.9	7837925.6
7	Estância Caiman	567961.2	7803860.3
8	Vera Lúcia	595015.0	7838415.5
9	Eldorado do Rio Negro	619640.5	7827305.9
10	Santa Sofia	562544.9	7826404.6
11	Primavera	614552.9	7843087.0

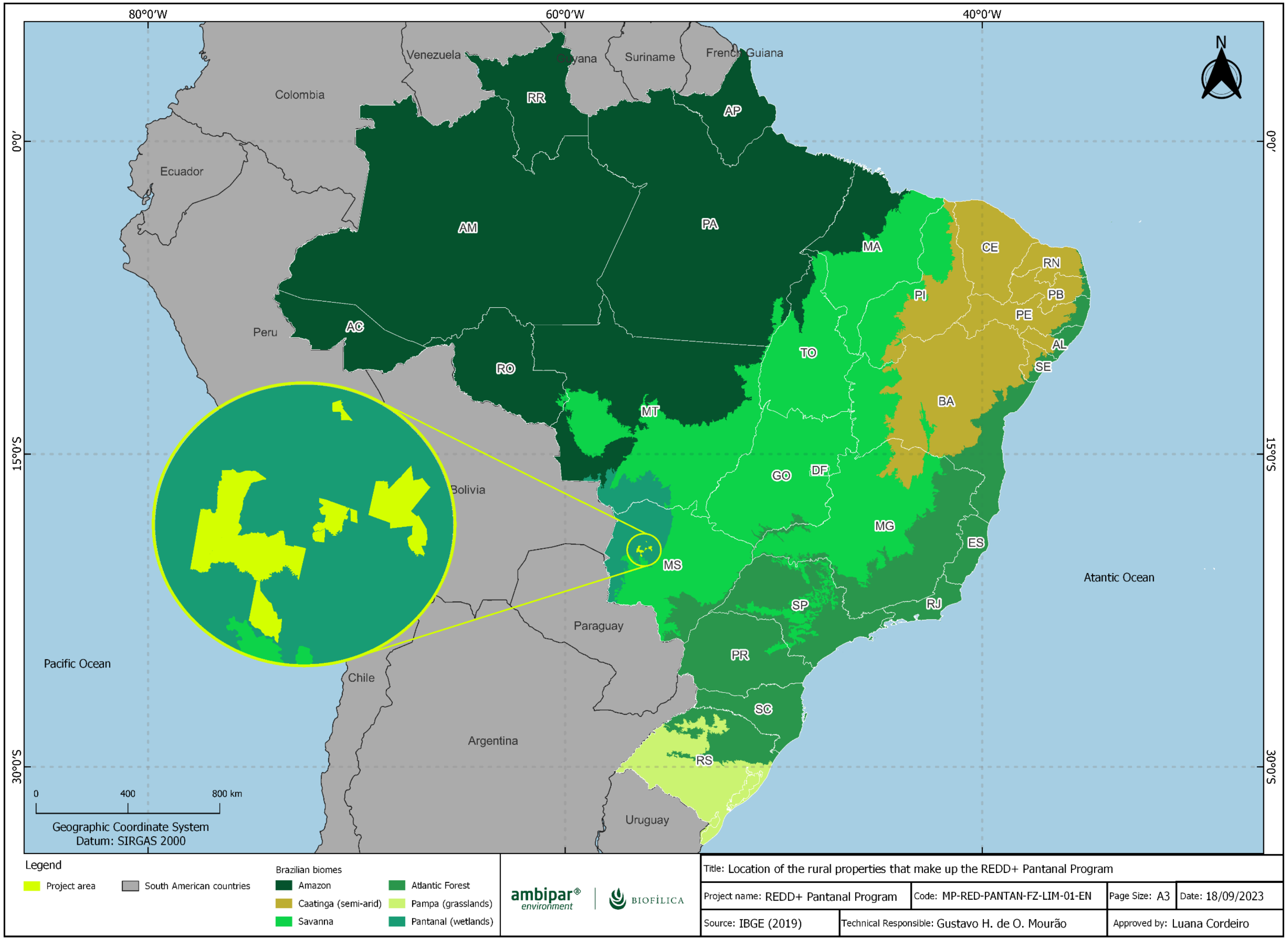

Figure 1. Location of the Pantanal biome, indicating the area where the Pantanal REDD+ Program is located

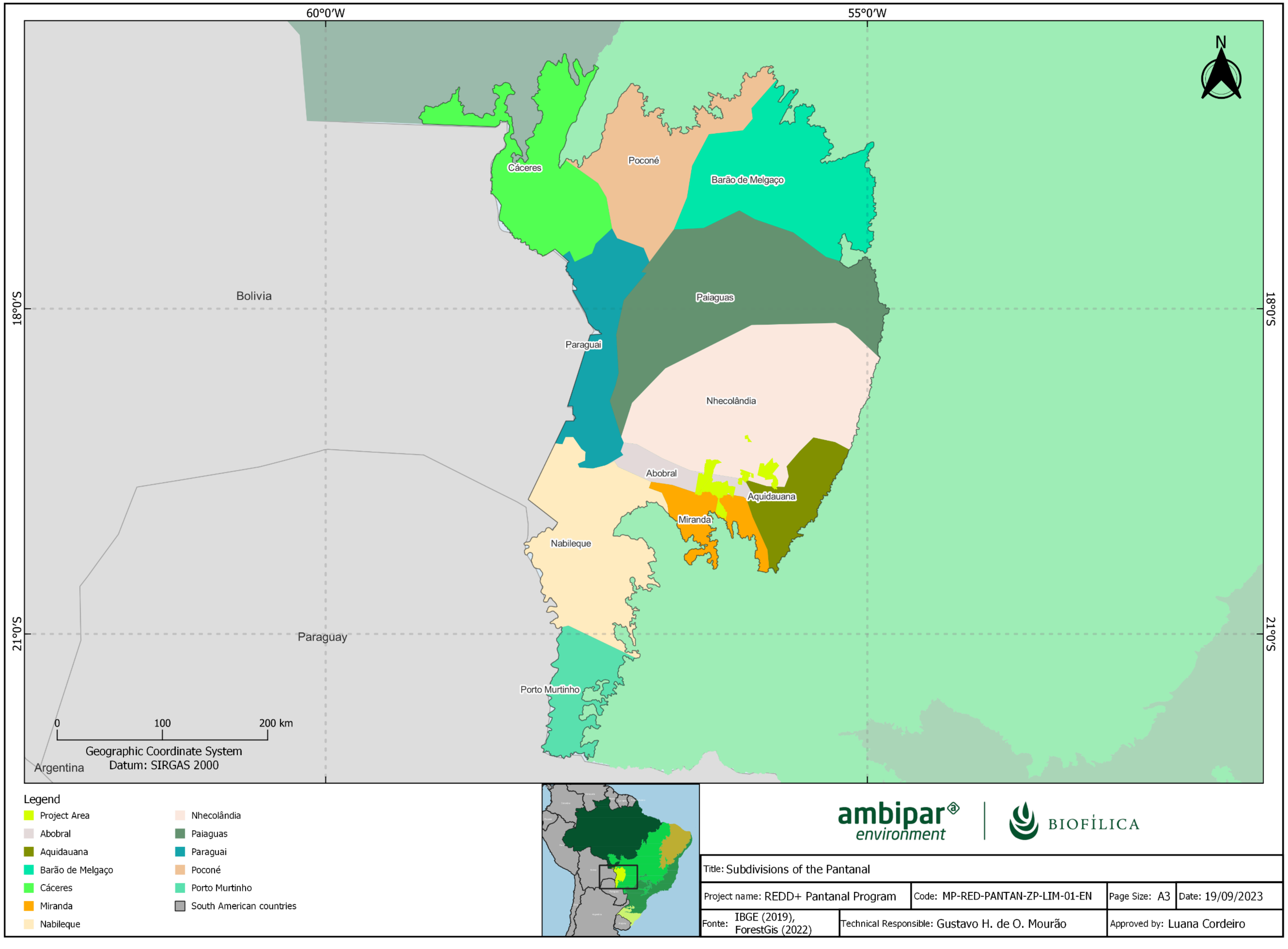

Figure 2. Location of the Pantanal biome and its 11 sub-regions, indicating the insertion area of the Pantanal REDD+ Program

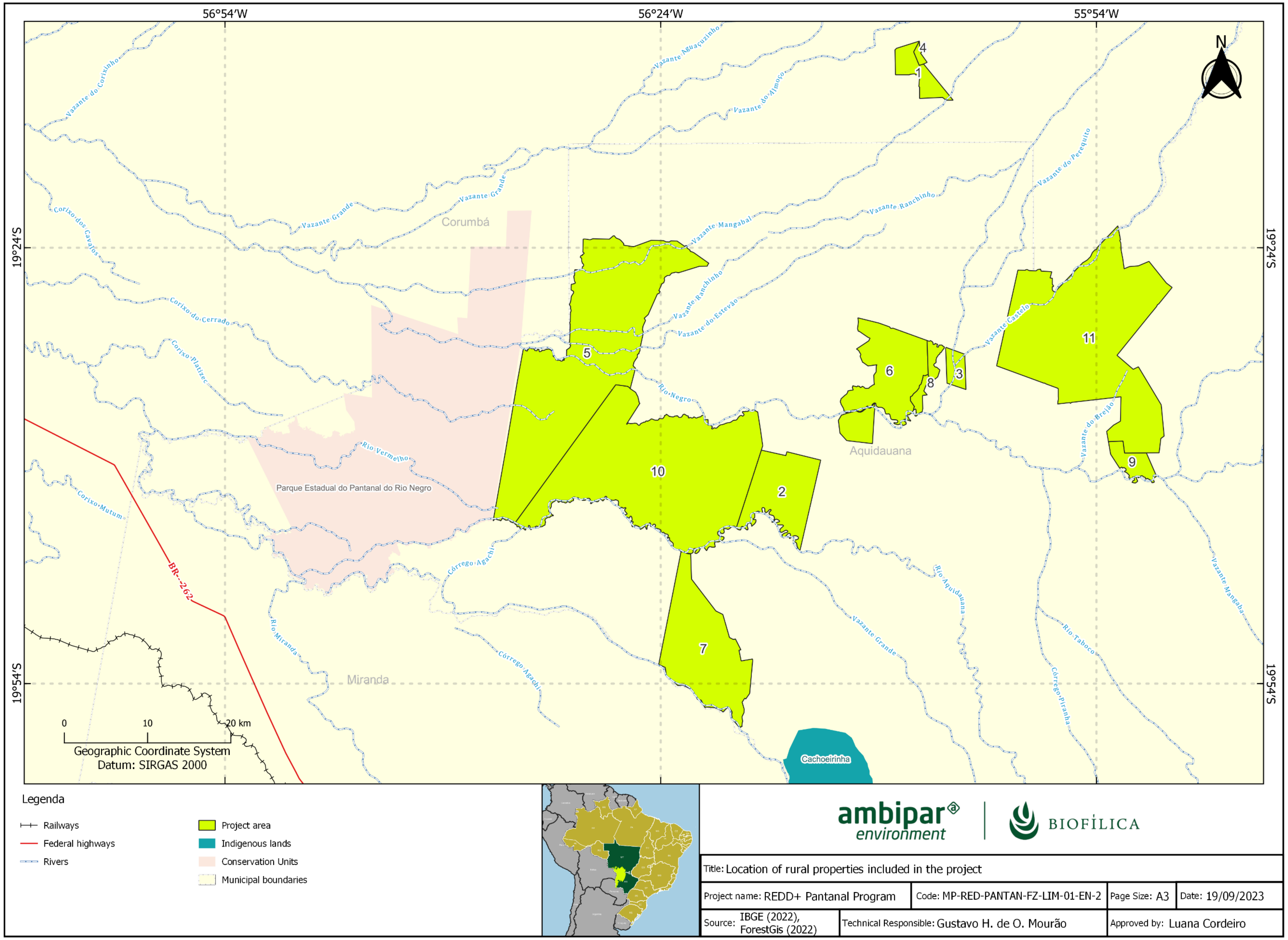

Figure 3. Location of the properties participating in the first instance of the Pantanal REDD+ Program

1.14 Conditions Prior to Project Initiation

The history of land use in the region where the Pantanal REDD+ Program is located is mainly geared towards livestock practices, with areas with forest cover being converted to pasture for animals. This deforestation dynamic in the Pantanal biome historically predates the date of implementation of the Pantanal REDD+ Program and, in this sense, this Program was not implemented with the aim of generating Greenhouse Gas emissions, but rather to transform the recurring dynamic that causes this fact, acting to prevent legalized deforestation inside private properties.

It is worth mentioning that the conditions prior to the start of the Pantanal REDD+ Program correspond to the baseline scenario. In this regard, below is a summary of the conditions prior to the start of the project, further details of which can be found in section 3.4 Baseline scenario.

- **Ecosystem type:** Flood and drought system, considered the largest floodplain in the world. It has a topography formed by mountains and plateaus, a tropical climate with rainfall well distributed throughout the year, fertile, sandy and clayey soil. It has the second largest biodiversity in Brazil, with the largest native vegetation cover among the country's biomes, and a concentration of endangered species.

- **Current and historical land use:** Historical local land use is predominantly dedicated to cattle breeding, which is a common practice in the Pantanal biome. Currently, land use in the area where the Pantanal REDD+ Program is being implemented is covered by forest and savannah.

- **Current and previous environmental conditions in the project area***:*

  Considering the geographical area of the Pantanal REDD+ Program (which includes the Pantanal biome), we can observe the following physical characteristics:

  Climate: The climate is hot and humid in the Pantanal region and, according to the Köppen classification system, the area falls into the Am, Af, and Aw classifications, each representing specific subtypes of the tropical climate with unique characteristics.

  Hydrology: The geographical area falls within the Paraguay Hydrographic Region (RH).

  Topography: The Pantanal region is an alluvial plain affected by rivers that drain the Upper Paraguay river basin, forming an extensive flooded area. Its plain is slightly undulating with rare isolated elevations and many shallow depressions.

  Relevant historical conditions: The Pantanal biome is extremely important for the environment, being one of the most exuberant and diverse nature reserves on Earth. Because of its characteristics and importance, this biome is considered a World Natural Heritage Site and a Biosphere Reserve recognized by UNESCO.

  Soils: The geographical area has predominantly sandy soils, comprising the following soil types: Agrisols, Cambisols, Chernosols, Gleisols, Latosol, Neosol Planosol, Plinthosol and Vertisol.

Vegetation: The vegetation typologies present in the geographical area of the Pantanal REDD+ Program are summarized as savannah typologies, but also include seasonal forests.

Ecosystems of the project area: The ecosystems of the Pantanal biome are characterized by savannahs and savannahs without periodic flooding, flooded fields and aquatic environments (freshwater or brackish lagoons, rivers, ebb and flow)[2].

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Under review.

The objective of the REDD+ Pantanal Program is in line with all current international, national, state and municipal legislation. The main laws and decrees that regulate the project's activities are listed and detailed below.

Theme: Brazilian Federal Constitution of 1988

It defines and regulates how laws should be created in the country, how the three branches of government (Executive, Legislative and Judiciary) should work and the bodies that act in conjunction with them. The Constitution's fundamental principles are*:*

I - sovereignty;

II - citizenship*;*

III - the dignity of the human person*;*

IV - the social values of work and free enterprise*;*

V - political pluralism*.*

Applicability to the project: Regulates and directs all national legislation that influences the project.

Theme: Land regulation

Related legislation*:*

- Law 601/1850 - Law of Vacant Lands: Provides for vacant lands in the Empire, and those that are owned by title of "sesmaria" without fulfilling the legal conditions, as well as by simple title of quiet and peaceful possession; and determines that, once the former have been measured and demarcated, they shall be ceded onerously, both to private companies and for the establishment of colonies of nationals and foreigners, the Government being authorized to promote foreign colonization in the manner stated.

[2] https://www.embrapa.br/pantanal/apresentacao/o-pantanal

- Law 4.504/1964 - Land Statute: Provides for the land statute and other measures. This law regulates the rights and obligations concerning rural real estate, for the purposes of implementing Agrarian Reform and promoting Agricultural Policy.
- Law 9.393/1996 - ITR Law - Tax on Rural Territorial Property: Provides for the ITR Tax, payment of the debt represented by Agrarian Debt Bonds and other provisions.
- Law 10.406/2002 - Brazilian Civil Code - About rights and duties in private life, of people, property and legal facts. Divided into two parts, (1) General Part, referring to people, property and legal facts and (2) Special Part, referring to the law of obligations, companies, things, family and successions.

Applicability to the project: The project complies with the laws listed, which guarantees the right to the property that includes the Project area and regularity with taxes related to rural properties.

Theme: Environment

Related legislation:

- Law 12.651/2012 - Brazilian Forest Code: Establishes general rules on the protection of vegetation, Permanent Preservation Areas and Legal Reserve areas; forest exploitation, the supply of forest raw materials, the control of the origin of forest products and the control and prevention of forest fires, and provides for economic and financial instruments to achieve its objectives.
- Law 9.605/1998 - Federal Law on Environmental Crimes: Provides for criminal and administrative sanctions arising from conduct and activities harmful to the environment and makes other provisions.
- MMA Ordinance No. 148, of June 7, 2022 - Recognizes the following official lists: "Official National List of Endangered Flora Species", "Official List of Endangered Brazilian Fauna Species" and "Official List of Extinct Brazilian Fauna Species".
- Law 7.347/1985 - Law on Public Civil Lawsuit: Regulates public civil lawsuit for liability for damage caused to the environment, the consumer, goods and rights of artistic, aesthetic, historical and tourist value and makes other provisions.
- Law No. 3.839, of December 28, 2009: Establishes the Mato Grosso do Sul State Territorial Management Program (PGT/MS); approves the First Approximation of the Ecological-Economic Zoning of the State of Mato Grosso do Sul (ZEE/MS), and makes other provisions.
- Law No. 5.235, of July 16, 2018: Provides for the State Policy for the Preservation of Environmental Services, creates the State Program for Payment for Environmental Services (PESA), and establishes a Management System for this Program;
- Law No. 4.555, of July 15, 2014: Establishes the State Policy on Climate Change - PEMC, within the territory of the State of Mato Grosso do Sul;
- Decree No. 15.798, of November 3, 2021: Regulates the Voluntary Public Registry of Annual Greenhouse Gas Emissions and the State Communication, provided in the State Climate Change Policy, provided in the State Law No. 4.555, of July 15, 2014;

- Law No. 4.163, of January 2, 2012: Regulates, within the scope of the state of Mato Grosso do Sul, the exploitation of forests and other forms of native vegetation, the use of forest raw materials, the obligation to replenish forests;
- Decree No. 14.755, of June 12, 2017: Provides for the institution and recognition of Private Natural Heritage Reserves, within the scope of the State of Mato Grosso do Sul;
- Decree No. 14.273, of October 8, 2015: Provides for the Restricted Use Area of the Pantanal floodplain, in the state of Mato Grosso do Sul;
- Decree No. 13.977, of June 5, 2014: Provides for the Rural Environmental Registry of Mato Grosso do Sul and the MS More Sustainable Program;
- SEMAC Resolution No. 11 of July 15, 2014: Implements and disciplines procedures relating to the Rural Environmental Registry and the MS More Sustainable Program referred to in State Decree No. 13,977 of June 5, 2014 (consolidated).
- Ordinary Law No. 2.548/2017: Provides for the Environmental Policy for the Protection, Control, Conservation and Recovery of the Environment in the Municipality of Aquidauana.
- Organic Law of the Municipality of Aquidauana-MS: Presents rules that regulate public life in the city, always respecting the Federal Constitution and the State Constitution.
- Organic Law of the Municipality of Miranda-MS: Presents rules that regulate public life in the city, always respecting the Federal Constitution and the State Constitution.
- Organic Law of the Municipality of Corumbá-MS: Presents rules that regulate public life in the city, always respecting the Federal Constitution and the State Constitution.

Applicability to the project: The project area includes Permanent Preservation Areas, Legal Reserve areas, and endangered species of Brazilian fauna and flora, which leaves the owner of the area susceptible to the consequences of the law if they fail to fulfill their responsibilities. In addition, the project falls within the Pantanal Floodplain Restricted Use Area, which has specific restrictions at state level.

It is worth mentioning here that Ambipar Group is committed to the environment, and its Code of Conduct and Compliance makes it explicit that all its activities and those of its employees must comply strictly with environmental legislation and standards, seeking to optimize natural resources.

Theme: Fighting corruption

Related legislation:

- Law 12.846/2013 - Anti-Corruption Law: Provides for the administrative and civil liability of legal entities for acts against the public administration, whether national or foreign, and makes other provisions.

Applicability to the project: It is important to emphasize that any action in this Program is contrary to acts of corruption or conduct aimed at personal gain to the detriment of any of the companies involved in the

Program, society or the Government, and it is incumbent on all those involved to refrain from such practices.

Theme: Labor laws

Related legislation:

- Decree-Law 5.452/1943 - Approves the Consolidation of Labor Laws (CLT), which describes all the rights and duties of employers and employees.

Applicability to the project: The REDD+ Pantanal Program is committed to complying with current labor laws and providing quality of life for its involved employees.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

The Pantanal REDD+ Program has not been registered with any other GHG emissions program. Currently, there is no national or international REDD+ regulatory regime applicable to the Pantanal REDD+ Program.

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

The Pantanal REDD+ Program has not been registered with any other GHG emissions program. Currently, there is no national or international REDD+ regulatory regime applicable to the Pantanal REDD+ Program.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

The REDD+ Pantanal Program has not been submitted for validation/verification under any other GHG program. Therefore, it has not been rejected by any other GHG program.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Not applied.

Brazil is not an Annex 1 country under the Kyoto Protocol and therefore has no commitments to reduce greenhouse gas emissions under the Convention. In addition, the Pantanal REDD+ Program does not have any projects related to carbon credit generation under the CDM or any other regulatory scheme within the project area.

As such, the project is not registered or seeking registration in any other program or mechanism that includes Emissions Trading and Other Binding Limits. Furthermore, the project does not have or intend to link any other form of GHG-related environmental credit.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Not applied.

The REDD+ Pantanal Program does not hold or wish to generate any type of environmental credit related to GHG emission reductions or removals claimed beyond the VCS Program.

1.17.3 Supply Chain (Scope 3) Emissions

Not applied.

It is not planned that the activities of the REDD+ Pantanal Program will have an impact on the emissions of goods and services in any supply chain.

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

All the activities carried out by the REDD+ Pantanal Program must follow the norms and rules established in the Code of Conduct and Compliance (2023), the Sustainability Policy (2023) and the Diversity and Inclusion Policy (2023) of the Ambipar Group, of which Biofílica Ambipar is a part.

In this sense, the REDD+ Pantanal Program, guided by the guidelines of the Ambipar Group's Code of Conduct and Compliance, is committed to sustainability and working to preserve and regenerate planet Earth. Its activities are based on the Sustainable Development Goals (SDGs) and reinforce its work with environmental, social and governance aspects, while its engagement with stakeholders brings it back to conscious capitalism, where the generation of value goes beyond its employees and seeks positive impacts on the environment, society and the global market.

As stipulated in the Ambipar Group's Sustainability Policy, the REDD+ Pantanal Program aims to establish principles, guidelines and responsibilities to be observed in all its processes, reinforcing its commitments to sustainable development and the circular and low-carbon economy. With regard to the farms that are members of the REDD+ Pantanal Program, they have various activities underway that contribute to the environmental conservation of the Pantanal biome and the sustainable development of the region.

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 2. Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	8.5	Achieve full and productive employment and decent work for all women and men, including young people.	Activities implemented to increase	The REDD+ Pantanal Program applies equal employment opportunities to all qualified people without any kind of discrimination, as well as equality, equity and universality of policies and initiatives to value diversity, with attention to minority groups. In addition, it is committed to hiring, inclusion and unconditional respect for all people, especially women, blacks, LGBTQIAPN+, people in situations of refuge, immigrants, people with disabilities and other minority groups.	Equal opportunities and remuneration for filling job vacancies generated as a result of the activities proposed for the Pantanal REDD+ Program.

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
2)	8.9	Draw up and implement policies to promote sustainable tourism, which creates jobs and promotes local culture and products.	Activities implemented to increase	Encourage properties that are members of the REDD+ Pantanal Program to carry out ecotourism activities.	Ecotourism activities generate positive impacts, such as: generating jobs; promoting Pantanal culture; raising awareness and preserving the Pantanal ecosystem; and raising funds that contribute to maintaining local biodiversity.
3)	10.2	Promote the social, economic and political inclusion of all, regardless of age, gender, disability, race, ethnicity, origin, religion, economic or other status.	Activities implemented to increase	In the development of the REDD+ Pantanal Program, discrimination of any kind is not tolerated on the grounds of gender, race, religion, sexual orientation, political opinion, origin, class or social condition, nationality, place of birth, age, pregnancy, health status, disability, genetic predisposition, lifestyle, union membership, retirement, disability, or any other individual characteristics.	Equal opportunities to fill job vacancies generated as a result of the activities proposed for the REDD+ Pantanal Program.

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
4)	12.b	Monitoring the impacts of sustainable development for sustainable tourism.	Activities implemented to increase	Monitor the properties that are members of the REDD+ Pantanal Program and adhere to ecotourism activities.	Ensuring that ecotourism activities generate positive impacts, such as: generating jobs; promoting Pantanal culture; raising awareness and preserving the Pantanal ecosystem; and raising funds that contribute to maintaining local biodiversity.
5)	13.0	Tons of greenhouse gas emissions avoided.	Activities implemented to increase	The REDD+ Pantanal Program aims to prevent the release of 1,887,177 tCO2e into the atmosphere during the first reference period (10 years).	It is estimated that the release of 1,887,177 tCO2e will be avoided over the lifetime of the Program.
6)	13.3	Improve education, raise awareness and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning.	Activities implemented to increase	The program plans to disseminate knowledge, instructions and experiences aimed at mitigating, adapting to, reducing the impact of and warning about climate change.	Holding workshops with local communities and other interested parties.

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
7)	15.1	Number of hectares of forest loss reduced in the project area measured against the without-project scenario	Activities implemented to increase	The Pantanal REDD+ Program has contributed to the conservation of local biodiversity and the maintenance of ecosystem services through the conservation of the Project Area by preventing the deforestation of 23,269 hectares of forest and forested savannah phyto physiognomies.	Conserved 23,269 hectares of forest and forested savannah that would have been cleared for land use conversion in the without-project scenario.
8)	15.2	Implement sustainable management and stop deforestation.	Activities implemented to increase	The REDD+ Pantanal Program plans to implement sustainable management of the forests within the project boundaries, with the aim of halting planned deforestation within the participating properties.	Propose conservation-oriented activities, adding value to 23,269 hectares of forest and forested savannah that would be cleared for land-use conversion in the without-project scenario.
9)	15.5	Red List Index	Activities implemented to increase	A biodiversity survey is being carried out in the project area, with the aim of ascertaining the presence and number of endangered species.	The aim is to implement activities to monitor biodiversity and species under some degree of threat, as well as promoting the maintenance of their habitats.

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
10)	15.a	Mobilize and significantly increase, from all sources, financial resources for the conservation and sustainable use of biodiversity and ecosystems.	Activities implemented to increase	The REDD+ Pantanal Program obtains economic incentives from the voluntary carbon market, through Reducing Emissions from Deforestation and Forest Degradation (REDD) through Avoided Planned Deforestation (APD).	The expected impact of the Program's activities is that the financial investments raised will result in improvements for everyone involved in the project. These efforts contribute to healthy ecosystems, climate change mitigation and the well-being of local communities.
11)	17.17	Effective public, public-private and civil society partnerships, based on experience of the strategies for mobilizing resources from these partnerships.	Activities implemented to increase	The REDD+ Pantanal Program already has strategic partnerships in place with the Taquari Vivo Institute and the 5P Alliance Association, and aims to formalize other strategic partnerships that are necessary to maintain and carry out the program's activities.	These partnerships promote more efficient use of resources, greater innovation and better results for the activities proposed by the project.

1.19 Additional Information Relevant to the Project

Leakage Management

Under development.

Commercially Sensitive Information

Under development.

Further Information

Under development.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	**Under development.**
Legal or customary tenure/access rights	Under development.
Stakeholder diversity and changes over time	Under development.
Expected changes in well-being	Under development.
Location of stakeholders	Under development.
Location of resources	Under development.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	Under development.
Stakeholder engagement process	Under development.
Consultation outcome	Under development.
Ongoing communication	Under development.
Stakeholder input	Under development.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Under development.
Outcome of FPIC	Under development.

2.1.4 Grievance Redress Procedure

Development process	Under development.
Grievance redress procedure	Under development.

Grievances received	Resolution and outcome
Under development.	Under development.
	…

2.1.5 Public Comments

Comments received	Actions taken
Under development.	Under development.
…	….

2.2 Risks to Stakeholders and the Environment

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	Under development.	Under development.
Working conditions	Under development.	Under development.
Safety of women and girls	Under development.	Under development.
Safety of minority and marginalized groups, including children	Under development.	Under development.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	Under development.	Under development.

2.3 Respect for Human Rights and Equity

All the activities carried out by the REDD+ Pantanal Program must follow the norms and rules established in the Code of Conduct and Compliance (2023), the Sustainability Policy (2023) and the Diversity and Inclusion Policy (2023) of the Ambipar Group, of which Biofílica Ambipar is a part.

The Code of Conduct and Compliance (2023) is an instrument for implementing the values and its main objective is to guide the conduct of all parties involved in adopting good practices in relationships and in carrying out activities.

In addition, the Ambipar Group also has a Sustainability Policy (2023) and a Diversity and Inclusion Policy (2023), which provide transparency and reinforce commitments to sustainable development and favor the provision of a favorable and healthy work environment for all professionals involved in its activities, valuing diversity and ensuring equal opportunities without any distinction as to the characteristics, history, religion or identity of individuals.

The REDD+ Pantanal Program is committed to inclusion and unconditional respect for all people, especially women, blacks, LGBTQIAPN+, people in refugee situations, immigrants, people with disabilities and other minority groups.

In the development of the Program, discrimination of any kind will not be tolerated on the grounds of gender, race, religion, sexual orientation, political opinion, origin, class or social condition, nationality, place of birth, age, pregnancy, state of health, disability, genetic predisposition, lifestyle, union membership, retirement, disability, or any other individual characteristics.

In addition, the REDD+ Pantanal Program expresses its rejection of any forced labor or labor in conditions analogous to slavery, any form of discrimination and harassment, both moral and sexual. Working conditions in the Program are ensured from the outset of the negotiation process, since before signing a contract with the owners of the farms participating in this Program, they must request a Labor Debt Clearance Certificate issued by the Labor Court, in order to confirm that the properties are in order in this regard.

2.3.1 Labor and Work

Discrimination and sexual harassment	Under development.
Management experience	Under development.
Gender equity in labor and work	Under development.
Human trafficking, forced labor, and child labor	Under development.

2.3.2 Human Rights

Guided by the Ambipar Group's Code of Conduct, the REDD+ Pantanal Program is committed to the communities around it, acting on the following premises:

- Encourage an attitude of respect for people, their traditions and values;
- Adopt and maintain a transparent process for defining social actions;
- Respect the interests and needs of the communities in which it operates;
- Collaborate with knowledge and technical information that can bring public benefits;
- Preserving and guaranteeing employees' political freedom and freedom of expression;
- The relationship of representatives of the REDD+ Pantanal Program with authorities, politicians and public agents should be based on transparent, professional and correct attitudes, and any form of pressure or request from public agents that does not correspond to this definition should be immediately rejected and reported to the management of Biofílica Ambipar;
- The relationship must be guided by respect for the laws and conventions that deal with fundamental human rights and the protection of sustainability. In all their dealings, representatives of the REDD+ Pantanal Program must observe ethical standards, health, safety and respect for human rights and social and environmental responsibility.

2.3.3 Indigenous Peoples and Cultural Heritage

Under development.

2.3.4 Property Rights

Rights to territories and resources	Under development.
Respect for property rights	Under development.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Under development.
Summary of the benefit sharing plan	Under development.
Approval and dissemination of benefit sharing plan	Under development.
Benefit sharing during the monitoring period	Under development.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	Under development.	Under development.
Soil degradation and soil erosion	Under development.	Under development.
Water consumption and stress	Under development.	Under development.
Usage of fertilizers	Under development.	Under development.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes ☐ No

Species and habitat	Under development.
…	…

2.4.2 Introduction of species

Under development.

2.4.3 Ecosystem conversion

Not applied.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF)	1.6
Module	VMD0001	Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP-AB)	1.1
Module	VMD0006	Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL)	1.3
Module	VMD0009	Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and avoiding planned wetland degradation (LK-ASP)	1.3
Module	VMD0013	Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB)	1.2
Module	VMD0015	Methods for monitoring of greenhouse gas emissions and removals (M-REDD)	2.2
Module	VMD0016	Methods for stratification of the project area (X-STR)	1.2

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Module	VMD0017	Estimation of uncertainty for REDD+ project activities (X-UNC)	2.2
Tool	VT0001	Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities (T-ADD)	3.0
Tool		Tool for AFOLU non-permanence risk analysis and buffer determination (T-BAR)	

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0007	The land in the project area must qualify as forest (according to the definition used by the VCS) for at least 10 years prior to the project start date.	According to an analysis of satellite images and information from the owners of the farms that are part of this program, the site where the conservation actions are being implemented has been classified as forest for more than 10 years.
VM0007	The baseline for deforestation and forest degradation within the project area must fall into one or more of the following categories: • Unplanned deforestation (VCS AUDD category)) • Planned deforestation/degradation (VCS APD category) Degradation due to extraction of wood for fuel (production of firewood and charcoal) (AUDD category of the VCS)	The Pantanal REDD+ Program falls under the category of avoided planned deforestation (VCS APD Category).

Methodology ID	Applicability condition	Justification of compliance
VM0007	Project activities to prevent planned deforestation can only be carried out in areas where there is legal permission to clear.	Project activities to prevent planned deforestation can only be carried out in areas where there is legal permission to clear.

3.3 Project Boundary

Under development.

3.4 Baseline Scenario

3.4.1 Conditions prior to project start

Below is a description of some of the environmental characteristics of the insertion region of the Pantanal REDD+ Program, collected through updated secondary data obtained from a literature review of the Pantanal biome and recent studies carried out in the implementation area of the Program.

Geology

According to the geological mapping of the Companhia de Pesquisa de Recursos Minerais (CPRM), made available through the Serviço Geológico do Brasil (SGB), the lithostratigraphy is divided into different hierarchies, as shown in Table 3 and the pedological map in Figure 4.

Table 3. Geology of the REDD+ Pantanal Program Geographical Area

Nome	Hierarquia	Sigla	Litotipos
Detrital-lateritic covers with ferruginous concretions	Not defined	ENdl	Sand, Clay, Gravel, Laterite
Ferruginous detrital-lateritic covers	Not defined	N1dl	Agglomerate, Sand, Clay, Laterite, Silt
Rio Apa Complex	Complex	PP3ap	Biotite gneiss, Cataclasite, Aluminous gneiss, Hornblende gneiss, Migmatite, Milonite, Granitic orthogneiss, Quartzite, Schist
Corpo Gabro Morro do Triunfo	Body	PP4δmt	Olivine gabbro, Troctolite
Granite Aluminum Body	Body	PP3γal	Granophyre, Leucogranite, Microgranite, Monzogranite, Syenogranite
Coxim Granite Body	Body	NP3γ4c	Granite, Tonalite

Nome	Hierarquia	Sigla	Litotipos
Rio Negro Granite Body	Body	NP3γ4n	Tonalito
São Vicente Granite Body	Body	C_cut_3γv	Adamelite, Granite
Taboco granite body	Body	NP3γ4t	Granitoid
Alluvial deposits	Not defined	Q2a	Sand, Arcose sand, Clay, Gravel, Polymictic conglomerate, Silt
Pantanal facies, alluvial deposits	Facies	Q1p2	Sand, Clay, Silt
Pantanal facies, colluvial deposits	Facies	Q1pc	Agglomerate, Sand, Clay, Silt
Pantanal facies, alluvial terraces	Facies	Q1p1	Sand, Clay, Laterite, Detrital-lateritic sediment, Silt
Alto Tererê Formation	Formation	PP4at	Amphibolite, Slate, Phyllite, Garnet-mica schist, Metarenite, Muscovite-quartz schist, Quartzite, Quartz schist, Metavolcanic rock
Aquidauana Formation	Formation	C2P1a	Sandstone, Diamictite, Shale, Siltstone
Bocaina Formation	Formation	NP3bo	Calcitic limestone, Dolomitic limestone, Dolomite
Cerradinho Formation	Formation	NP3ce	Arcosseous, Sandstone, Arcosean Sandstone, Argillite, Conglomerate, Dolomite, Shale, Marl, Siltstone
Training Coimbra	Formation	Sc	Sandstone, Conglomerate
Diamantino Formation	Formation	NP3di	Archosaic, Argillite, Shale, Siltstone
Furnas Formation	Formation	D1f	Arcosseous, Sandstone, Conglomerate, Siltstone, Subarcosseous
Jauru Formation	Formation	C1ja	Sandstone, Conglomerate supported by matrix, Diamictite, Shale, Sandy siltstone
Nobles Formation	Formation	NP3arn	Sandstone, Argillite, Dolomite, Silexite, Siltstone
Puga Formation	Formation	NP3pu	Sandstone, Diamictite, Mudstone, Metaconglomerate, Metadiamictite
Raizama Formation	Formation	NP3ra	Archosaic, Sandstone, Argillite, Conglomerate, Siltstone

Nome	Hierarquia	Sigla	Litotipos
Ronuro Formation	Formation	N1r	Sand, Clay, Gravel
Santa Cruz Formation	Formation	NPcz	Arcosseous, Sandstone, Conglomerate, Jaspelite
Tamengo Formation	Formation	NP3t	Slate, Sandstone, Cataclastic breccia, Calcareous limestone, Dolomitic limestone, Dolomite, Shale, Pelite
Urucum Formation	Formation	NPu	Sandstone, Conglomerate, Grauvaca, Siltstone
Xaraiés Formation	Formation	Q2x	Limestone tuff
Aguapeí Group	Group	MP3a	Slate, Oligomitic conglomerate, Phyllite, Clast-supported metaconglomerate, Quartz metarenite, Metasiltite
Alto Jauru Group	Group	PP4aj	Komatiitic basalt, Banded iron formation, Metabasalt, Metatuff, Shale
Pontes e Lacerda Group	Group	MP1pl	Phyllite, Metabasalt, Metatuff, Ferruginous quartzite, Schist
Mimoso Volcanic Sequence	Sequence	C-cut_3αm	Dacito, Riodacito, Riolito
Fecho dos Morros Suite	Suite	TJ_lambda_fm	Andesite, Latite, Nepheline syenite, Trachyte
Cachoeirinha intrusive suite	Intrusive suite	MP1γc	Granodiorite, Monzogranite, Tonalite
Ponta do Morro Suite	Suite	K2γ4pm	Biotite granite, Alkaline granite
Serra da Bocaina Suite	Suite	PP3αsb	Pyroclastic breccia, Dacite, Riodacite, Riolite, Lapillitic tuff
Cuiabá Unit - Subunit 1	Unit	NPcu1	Phyllite, Metarenite
Cuiabá Unit - Subunit 2	Unit	NPcu2	Calcareous marble, Metarosseous, Metarenite, Arcosean metarenite
Cuiabá Unit - Subunit 3	Unit	NPcu3	Phyllite, Itabirite, Calcitic marble, Metaconglomerate, Metarosseous, Metarenite
Cuiabá Unit - Subunit 5	Unit	NPcu5	Phyllite, Metaconglomerate, Metasediment, Metarenite
Cuiabá Unit - Subunit 6	Unit	NPcu6	Phyllite, Metarenite
Cuiabá Unit - Indivisible Subunit	Unit	NPcui	Phyllite, Metarenite

The project's main lithostratigraphies are Pantanal facies, alluvial deposits (Q1p2), Alluvial deposits (Q2a) and Pantanal facies, alluvial terraces (Q1p1), which together account for 90% of the geographical area of the Pantanal REDD+ Program. Below is a brief description of each of these lithostratigraphies.

- Pantanal facies, alluvial deposits (Q1p2):

  They comprise the upper portion, known as the alluvial deposits facies, made up of sub-recent clay-silty-sandy sediments (LACERDA FILHO *et al.*, 2004). It makes up 68% of the geographical area, making it the largest division of lithostratigraphy in the region of interest.

- Alluvial deposits (Q2a):

  These are deposits characterized by unconsolidated sediments, dominantly sandy, represented by sands with levels of gravel and lenses of silty-clayey material. They occur associated with the channels of larger watercourses, embedded in both the crystalline basement and the Tertiary deposits, basically comprising alluvial sediments. The holocene fluvial sedimentation pattern of these drainages is generally characterized by lateral accretion deposits of channel margin and bottom load, which include point bars, mid-channel bars and bottom load deposits. These sediments are also distributed in the flood plains of rivers where there is a lacustrine environment, represented by residual lakes, formed by the migration of the lateral accretion ridges of the bars, as well as dammed lakes. The probable age of these deposits is Pleistocene, obtained from the fossiliferous content found in the alluvium and alluvial paleoterraces of some of the region's rivers. Associated with these sediments, important concentrations of gold and diamond are found in the region (LACERDA FILHO et al., 2004). The Alluvial Deposits cover approximately 17% of the geographical area.

- Pantanal facies, alluvial terraces (Q1p1):

  They represent the intermediate portion, comprising elevated alluvial terrace facies, characterized as an old alluvial plain, encompassing sandy-clayey, partially unconsolidated and laterized sediments (LACERDA FILHO et al., 2004). Pantanal facies, alluvial terraces correspond to 5% of the geographical area.

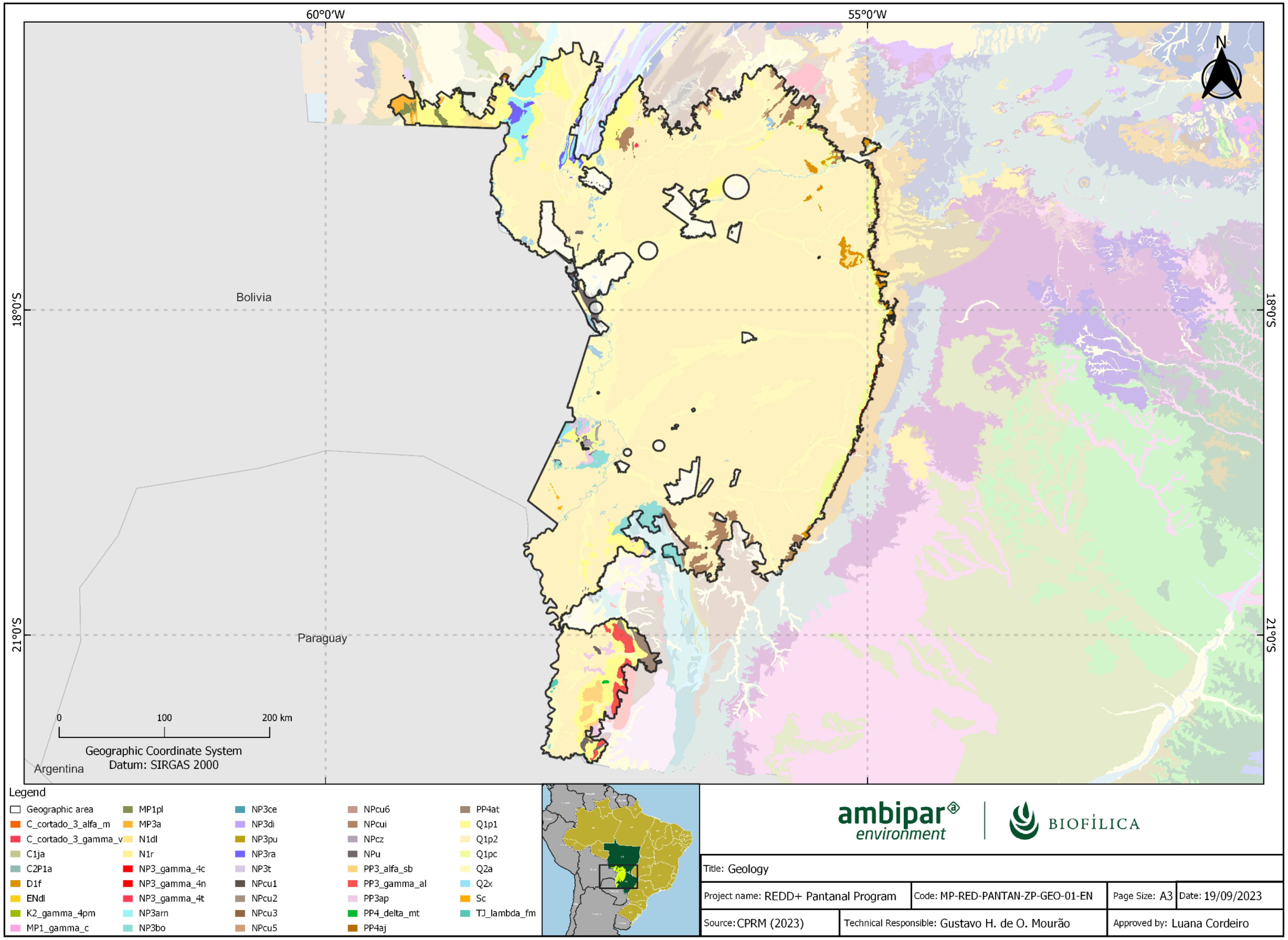

Figure 4. Geology of the geographical area of the Pantanal REDD+ Program

Geomorphology

The basic principle of geomorphological analysis is the ordering of geomorphological facts according to a temporal and spatial classification, with the modeled areas as the basic unit and their hierarchically related groups. Following a decreasing order of hierarchical magnitude, the following taxa are identified: Morphostructural Domains, Geomorphological Regions, Geomorphological Units, Modeled and Symbolized Landforms.

As mentioned above, in descending order of magnitude, comprising the largest taxa in the compartmentalization of the landform, the Morphostructural Domains occur on a regional scale and organize the geomorphological facts according to the geological framework marked by the nature of the rocks and the tectonics that act on them. This, combined with the effects caused by climate variation over geological time, form extensive groups of landforms with unique characteristics whose features, although diverse, maintain common relationships with the geological structure from which they were formed (IBGE, 2009).

According to the Geomorphology Technical Manual (IBGE, 2009), new morphostructural concepts classify four domains for the whole of Brazil, as shown in Figure 5: Phanerozoic Sedimentary Basins and Covers, Neoproterozoic Mobile Belts, Neoproterozoic Cratons, and finally, the main morphostructural domain where the REDD+ Pantanal Program area is located is called Quaternary Sedimentary Deposits. Below is a brief description of these domains.

- Phanerozoic Sedimentary Basins and Covers:

They comprise highlands and plateaus developed on horizontal to sub-horizontal sedimentary rocks, eventually folded and/or faulted, in different sedimentation environments, arranged on the continental margins and/or in the interior of the continent (IBGE, 2009). In the Geographical Area of the Pantanal REDD+ Program, the Phanerozoic Sedimentary Basins and Covers correspond to 1.15% of the total area.

- Neoproterozoic mobile belts:

They comprise extensive areas represented by plateaus, mountain alignments and interplanaltic depressions made up of folded and faulted terrain, mainly including metamorphites and associated granitoids (IBGE, 2009). In the Geographical Area, the Neoproterozoic Mobile Belts account for 8% of the total area.

- Neoproterozoic cratons:

Residual plateaus, plateaus and interplanaltic depressions, based on metamorphites and associated granitoids and covering sedimentary rocks and/or volcano-plutonism, deformed or not (IBGE, 2009). In the Geographical Area, the Neoproterozoic Cratons correspond to 1.63% of the total area.

- Quaternary sedimentary deposits:

This domain is made up of accumulation areas represented by low slope plains and terraces and, eventually, depressions modeled on horizontal to sub-horizontal sediment deposits from fluvial, marine, fluvial-marine, lagoon and/or aeolian environments, arranged in the coastal zone or inland from the continent (IBGE, 2009). In the Geographical Area, Quaternary Sedimentary Deposits predominate, accounting for 88% of the total area.

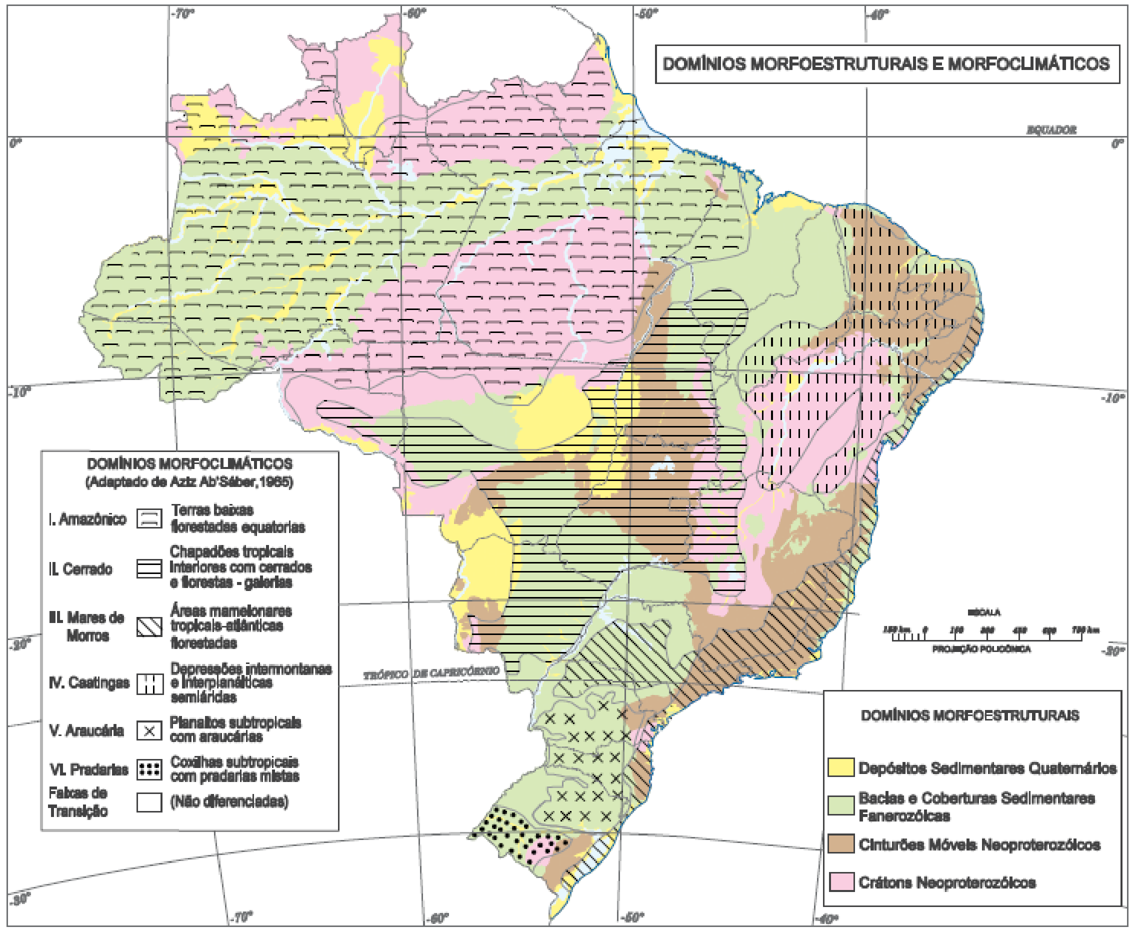

Figure 5. Morphostructural and morphoclimatic domains. IBGE, 2009

The second hierarchical level of relief classification is represented by Geomorphological Regions, which are compartments within litho morphostructural groups where, through the action of past and present climatic factors, common genetic characteristics are attributed, forming a set of similar features related to surface formations and phytophysiognomies. The third level is the Geomorphological Units, which are understood as an arrangement of altimetric and physiognomically similar forms in their various types of modeling. The modeled, which represent the fourth level, are patterns of relief forms that have a similar geometric definition (IBGE, 2009). The geomorphological summary of the study region is shown in Table 4 and Figure 6.

Table 4. Geomorphological overview of the REDD+ Pantanal Program Geographical Area

Morphostructural Domain	Geomorphological region	Geomorphological Units	Modeled	Forms of Relief
Phanerozoic Basins and Sedimentary Covers	Paraná Basin Residual Plateaus	Chapadão do Rio Corrente	Dt; Pgu; Pgi	tabular top; pediplane
	Plateaus and Levels of the Western Edge of the Paraná Basin	Dissected Plateau of the Western Edge of the Paraná Basin	Dt; Pri	tabular top; pediplane
	Plateaus of the Western Borders of the Paraná Basin	Chapada dos Guimarães	Dt; Pru; Pgi; Da	tabular top; pediplane; sharp top
	Plateaus and Levels of the Western Edge of the Paraná Basin	First Plateau of the Western Edge of the Paraná Basin	Pru; Dt; Pgi; Da; Dc; Pi	pediplane; tabular top; sharp top; convex top; plane of undifferentiated genesis
	Sandstone-Basalt Plateaus	Planalto doTaquari -- Itiquira	Arc; Dt; Pgi; Da	colluvial plane; tabular top; pediplane; sharp top
Neoproterozoic Mobile Belts	Bodoquena Plateau	Serra da Bodoquena	Kc; Da; Dc	karst; sharp top; convex top
	Mountainous Complexes of the Upper Paraguay Plateaus	Serras de São Vicente – Mimoso	Dei; Da; Pgi; Pri	steep erosion slope; sharp top; pediplane
		Província Serrana	Da; Pgu; Pru; Dc; Pri; Dei	sharp top; pediplane; convex top; steep erosion slope
	Bodoquena Plateau	Eastern Bodoquena Mountain Alignments	Dt; Dc	tabular top; convex top

Morphostructural Domain	Geomorphological region	Geomorphological Units	Modeled	Forms of Relief
	Mountains and Hills of Mato Grosso do Sul	Morrarias do Urucum – Amolar	Pgi; Dc; Pgu; Da; Dt	pediplane; convex top; sharp top; tabular top
	Depressions of the South of Mato Grosso	Northern Bodoquena Depression	Dc; Pri	convex top; pediplane
		Cuiabana Depression	Arc; Da; Pri; Dt; Dc; Pru	colluvial plane; sharp top; pediplane; tabular top; convex top
		Upper Paraguay Depression	Dt; Dc; Pri; Ai; Da	tabular top; convex top; pediplane; squat plane; sharp top
Neoproterozoic cratons	Unidentified region - domain 4	Dissected surface of Jauru	Dc; Dt	convex top; tabular top
	Southern Sertaneja Depression	Middle Paraguaçu pediplane	Aptf	plain and terrace
	Guaporé residual plateaus	Guaporé Residual Plateaus	Dc; Pgu; Dt; Arc; Dei; Pri; Da	convex top; pediplane; tabular top; colluvial plane; steep erosion slope; sharp top
	Guaporé Depression	Guaporé Depression	Pri; Arc; Dt	pediplane; colluvial plane; tabular top
	Mountains and Hills of the Lower Paraguay river	Western Bodoquena plateaus and residual ridges	Dc; Da	convex top; sharp top
		Western Bodoquena Mountains	Dt; Dc	tabular top; convex top
Quaternary sedimentary deposits	Mato Grosso's plains and wetlands	Pantanal do Uberaba – Mandioré	Apl; Apfl; Pri; Ai	plain; pediplane; lowered plane
		Pre-Pantanal Floodplains	Pri; Dt; Arc; Da	pediplane; tabular top; colluvial plane; sharp top
		Pantanal do Negro – Taboco	Apfl; Arc; Ai; Apf	plain; colluvial plane; lowered plane
		Miranda-Aquidauana Pantanal	Dt; Atf; Apfl; Apf; Ai; Aptf; Dc	tabular top; terrace; plain; flattened plane;

Morphostructural Domain	Geomorphological region	Geomorphological Units	Modeled	Forms of Relief
				plain and terrace; convex top
		Lower Paraguay Plains and Wetlands	Pri; Apf; Ai; Aptf	pediplane; planície; lowered plane; plain and terrace
		Corixo Grande wetland	Pri; Ai	pediplane; lowered plane
		Lower Taquari Pantanal -- Paraguay	Ouch	lowered plane
		Jacadigo-Nabileque Plains and Wetlands	Apl; Pri; Apf; Ai	plain; pediplane; lowered plane
		Poconé wetlands	Pri; Ai	pediplane; lowered plane
		Itiquira wetlands -- São Lourenço	Arc; Da; Pgi; Ai; Pri	colluvial plane; sharp top; pediplane; lowered plane
		Paraguay River Plains and Wetlands -- Cuiabá	Apfl; Atfl; Ai; Atf; Aptf; Apf	plain; terrace; flattened plane; plain and terrace
		Pantanal do Taquari -- Nhecolândia -- Paiagás	Apf; Apfl; Ai	plain; lowland

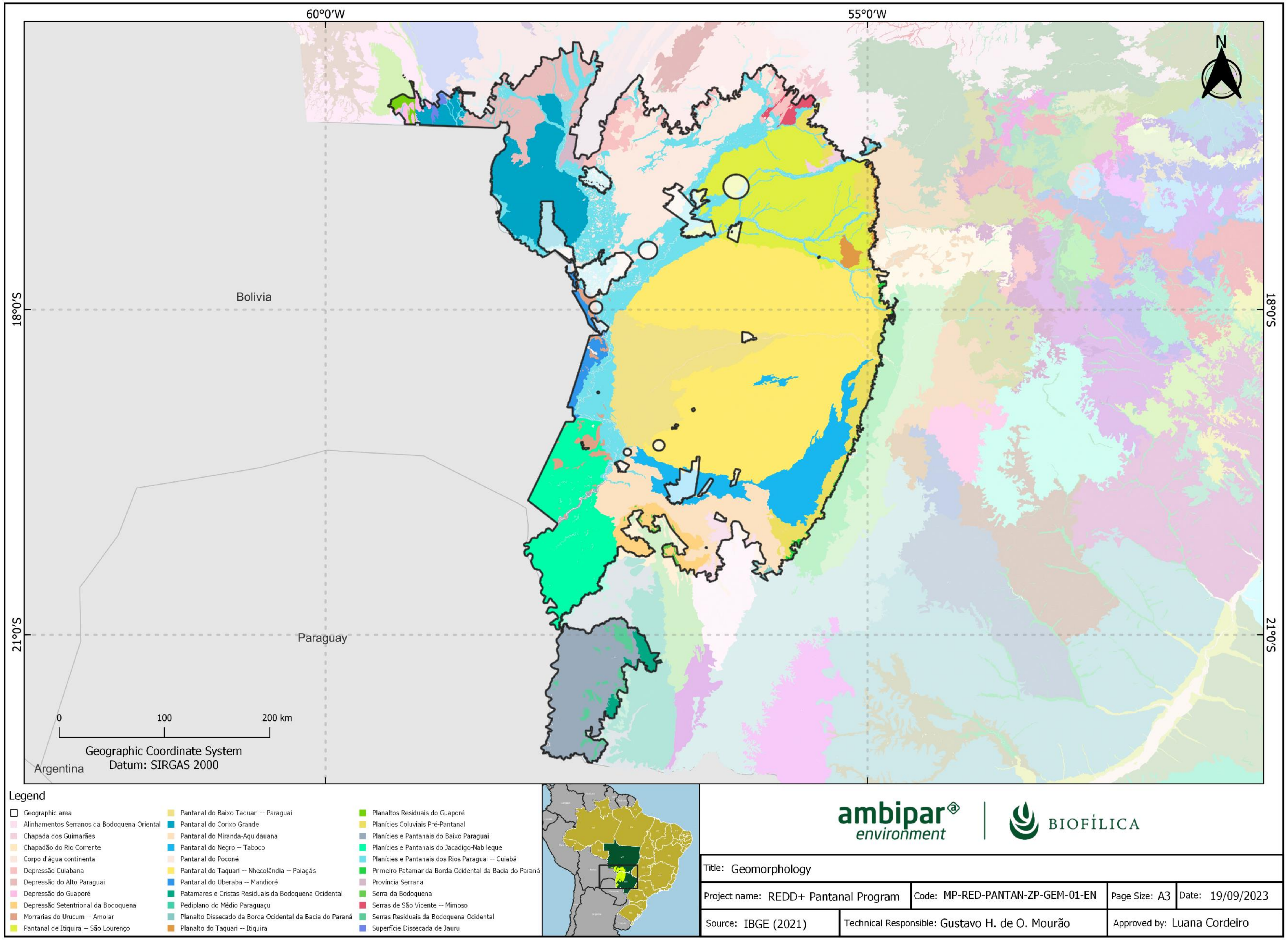

Figure 6. Geomorphology in the Geographical Area of the Pantanal REDD+ Program

Pedology

The Pedology presented in this section was drawn up based on the cartographic product made available by the Brazilian Institute of Geography and Statistics (IBGE), as well as its Technical Manual of Pedology, third edition, published in 2015 (IBGE, 2015). The geomorphological summary of the study region is shown in Table 5 and the pedological map is shown in Figure 7.

Table 5: Pedological overview of the Geographical Area of the Pantanal REDD+ Program

Order	Suborder	Large Group	Subgroup	Horizon
Argisol	Red-Yellow	Eutrophic	typical	A
	Red-Yellow	Dystrophic	typical	A
	Red	Eutrophic	typical	A
	Red	Dystrophic	typical	A
Cambisol	Habitat	Tb Dystrophic	typical	A
	Habitat	Tb Eutrophic	chernosolic	A
Chernosol	Habitat	Optical	typical	A
	Argillic	Optical	typical	A
	Rhinzic	Optical	typical	A
Gleisol	Melanic	Tb Dystrophic	typical	A
	Habitat	Tb Eutrophic	typical	A
	Habitat	Ta Eutrophic	typical	A
Latosol	Red-Yellow	Dystrophic	argisolic	A
	Yellow	Dystrophic	typical	A
	Red	Eutrophic	typical	A
	Red	Dystrophic	typical	A
Neosol	Quartzarenic	Optical	typical	A

Order	Suborder	Large Group	Subgroup	Horizon
	Quartzarenic	Hydromorphic	typical	A
	Fluvial	Ta Eutrophic	vertisolic and typical	A
	Litholic	Dystrophic	typical	A
	Regolithic	Dystrophic	typical	A
	Regolithic	Eutrophic	typical	A
	Fluvial	Tb Dystrophic	typical	A
	Litholic	Eutrophic	typical	A
	Litholic	Chernosolic	typical	A
	Red	Eutrophic	latosolic	A
Planosol	Nautical	Optical	typical	A
	Habitat	Dystrophic	typical	A
	Habitat	Eutrophic	typical	A
Plinthosol	Argillic	Dystrophic	typical	A
	Habitat	Dystrophic	typical	A
	Metric	Concrete	argisolic	A
	Argillic	Eutrophic	typical	A
Vertisol	Hydromorphic	Sodium	solodic	A
	Hydromorphic	Carbonate	typical	A
	Habitat	Optical	typical	A
	Ebonics	Optical	chernosolic	A

Order	Suborder	Large Group	Subgroup	Horizon
	Habitat	Carbonate	chernosolic	A
	Hydromorphic	Optical	typical	A

- Argisols:

This class derives its name from the Latin *argilla,* meaning soils with a clay accumulation process, and its main associated characteristic is the textural B horizon (Bt). Another striking feature of this class is the increase in clay from the superficial A horizon to the subsurface B horizon, usually accompanied by good color distinction and other characteristics. The colors of the A horizon are always darker and those of the Bt horizon vary from grayish to reddish. They vary in depth but are generally shallow and deep. Together with the Latosols, they are the most significant soils in Brazil (IBGE, 2015). In the Geographical Area, Argisols account for 3.37% of the total area.

- Cambisol:

Cambisols are characterized by their variability in terms of depth, ranging from shallow to deep. In terms of drainage, these soils can vary from marked to imperfect. A distinctive feature of Cambisols is that they can have any type of A horizon over an incipient B horizon (Bi), which can have various colors. In many cases, these soils are stony, gravelly or even rocky (IBGE, 2015). In the Geographical Area, Cambisols account for 0.55% of the total area.

- Chernosol:

Chernosols are soils characterized by a surface horizon rich in organic matter, resulting in dark colors and high fertility. Below this, the subsurface horizons have reddish or dark tones, with high-activity clay, indicating a remarkable capacity for nutrient retention and exchange, making them ideal for agricultural activities (IBGE, 2015). In the Geographical Area, Chernosols account for 0.36% of the total area.

- Gleisols:

This class of soil derives its name from the Russian *glay,* which means pasty soil mass, connoting excess water and its main associated characteristic is the glei horizon. They are soils with characteristics of flooded areas or areas prone to flooding, such as river banks, islands, large plains, among others. In general, their color is grayish, bluish or greenish, within 50 centimeters of the surface. It can have natural fertility ranging from high to low, and in conditions of poor drainage, its use is limited. This type of soil occurs in all Brazilian regions, but predominantly occupies the flood plains of rivers and streams (IBGE, 2015). In the Geographical Area, Gleisols represent 6.65% of its total area.

- Latosols:

This class of soil takes its name from the Latin lat, which means highly altered material (brick), connoting a high content of sesquioxides, and its main associated characteristic is the oxisolic B horizon. They are generally highly weathered, deep soils with good drainage. They are known for having great homogeneity of characteristics throughout the profile, the mineralogy of the clay fraction is predominantly kaolinitic or kaolinitic-oxic. They are well distributed in all Brazilian regions, differing in color and iron oxide content (IBGE, 2015). In the Geographical Area, yellow latosols predominate, corresponding to 34.89% of the total area, and latosols correspond to 2.36%.

- Neosols:

This class of soil takes its name from the Greek *néos*, which means new, modern, connoting young soils, at the beginning of formation, and its main associated characteristic is its small development. Neosols are made up of mineral or organic material that is not very thick, less than 30 centimeters thick, and do not have any type of B horizon (IBGE, 2015). In the Geographical Area, this type of soil accounts for 22.47% of its total area, making it the second most representative soil in the region.

- Nitosols:

This soil class derives its name from the Latin *nítidus*, meaning shiny, connoting shiny surfaces in structural units, and its main associated feature is the nitic B horizon. This B horizon is characterized by a subsurface horizon, with structural development of the prism or block type, varying from moderate to strong and with shiny aggregate surfaces, related to cerosity or compression surfaces. They have a clayey or very clayey texture, and the textural difference is inexpressive (IBGE, 2015). Red Nitosols occur in practically all of Brazil. In the Geographic Area, this type of soil corresponds to 0.18% of its total area.

- Planosols:

Planosols are mineral soils with imperfect or poor drainage. They have an eluvial surface or subsurface horizon with a lighter texture that contrasts sharply with the underlying B horizon, which is dense and rich in clay. This composition often results in a pan horizon, which retains a suspended water table periodically throughout the year. These soils can have different combinations of horizons, but they are marked by an accentuated textural change between the surface horizons (A and/or E) and the subsurface horizon (plannic). Their fertility varies, and in addition to texture, they show notable differences in structure, porosity and permeability between the horizons (IBGE, 2015). In the Geographical Area, Planosols account for 40.75% of the total area, and are the most representative soil in the region.

- Plinthosols:

They are mainly characterized by the presence of expressive plintitization with or without petroplinthite (iron concretions or yokes) (IBGE, 2015). In the Geographical Area, Plinthosols account for 17.43% of the total area, and are the third most representative soil in the region.

- Vertisols:

Vertisols are deep and shallow mineral soils with a vertic horizon, varying in color from dark to yellowish, grayish or reddish. They are characterized by the presence of cracks resulting from the expansion and contraction of the clay material, slickensides and a strongly developed prismatic structure. The typical sequence of horizons is A-Cv or A-Biv-C, without fitting the characteristics of Chernosols and without presenting lithic contact, petrocalcic horizon or duripan in the first 30cm (IBGE, 2015). They are fertile soils, found in climatic and relief conditions that prevent the removal of basic cations. Vertisols account for 4.54% of the total in the Geographical Area.

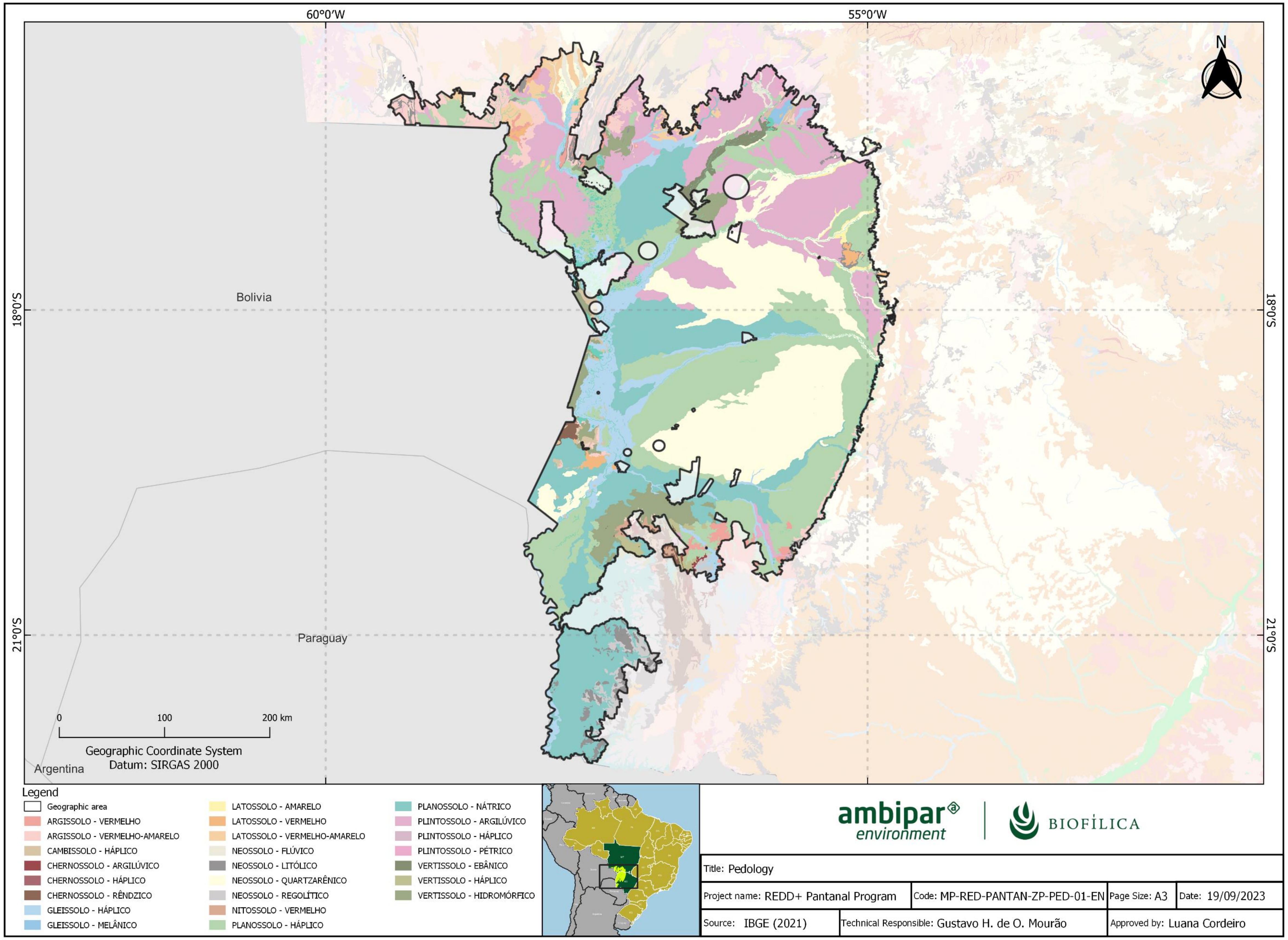

Figure 7. Pedology in the Geographical Area of the REDD+ Pantanal Program

Hydrography

The geographical area of the Pantanal REDD+ Program is located in the Paraguay Hydrographic Region (RH).

The Paraguay River Hydrographic Region (RH-Paraguay) represents the Brazilian portion of the international hydrographic basin of the Paraguay River. This region is strategically important for the management of Brazil's water resources, encompassing the Mato Grosso Pantanal. The Paraguay River, with a total length of 2,695 km (1,693 km within the RH-Paraguay), originates in the Chapada dos Parecis and is crucial for the drainage of the Pantanal plain. The RH-Paraguay covers 362,380 km², distributed between the states of Mato Grosso (48%) and Mato Grosso do Sul (52%), covering 86 municipalities and around 2.4 million inhabitants (ANA, 2023). Within the geographic area of the Program there are two Hydrographic Mesoregions (Upper Paraguay and Middle Paraguay) and eleven Hydrographic Microregions: Upper Guaporé, Upper Paraguay, Upper São Lourenço, Apa, Cuiabá, Middle Paraguay, Miranda, Negro (MS), Northwest Pantanal, Piquiri-Itiquira and Taquari. The RH, MeRH and MiRH are named after the main tributary that makes them up. Figures 8 and 9 show the hydrography map of the Geographical Area of the Pantanal REDD+ Program.

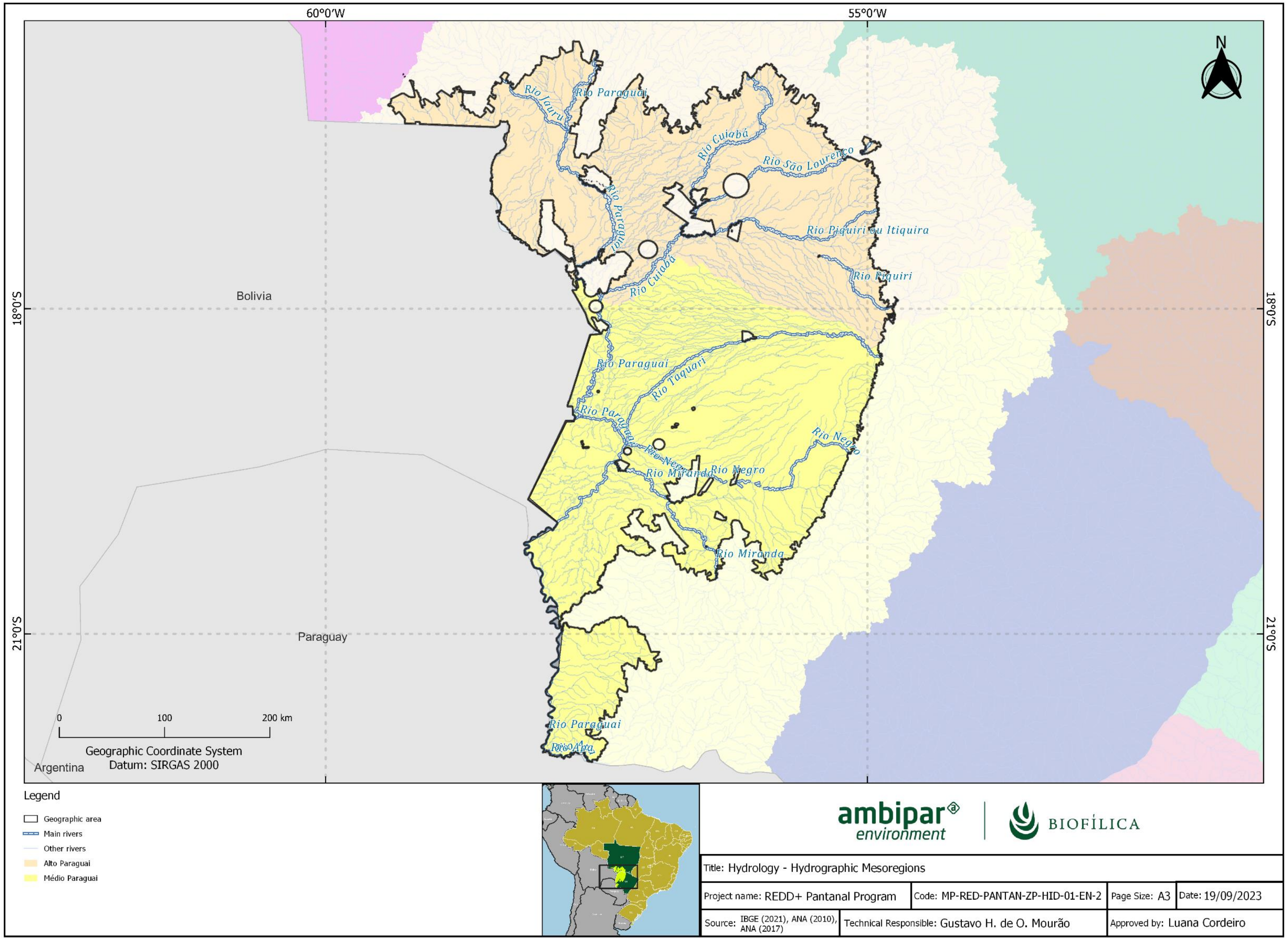

Figure 8. Hydrography and Hydrographic Regions of the Geographic Area of the Pantanal REDD+ Program

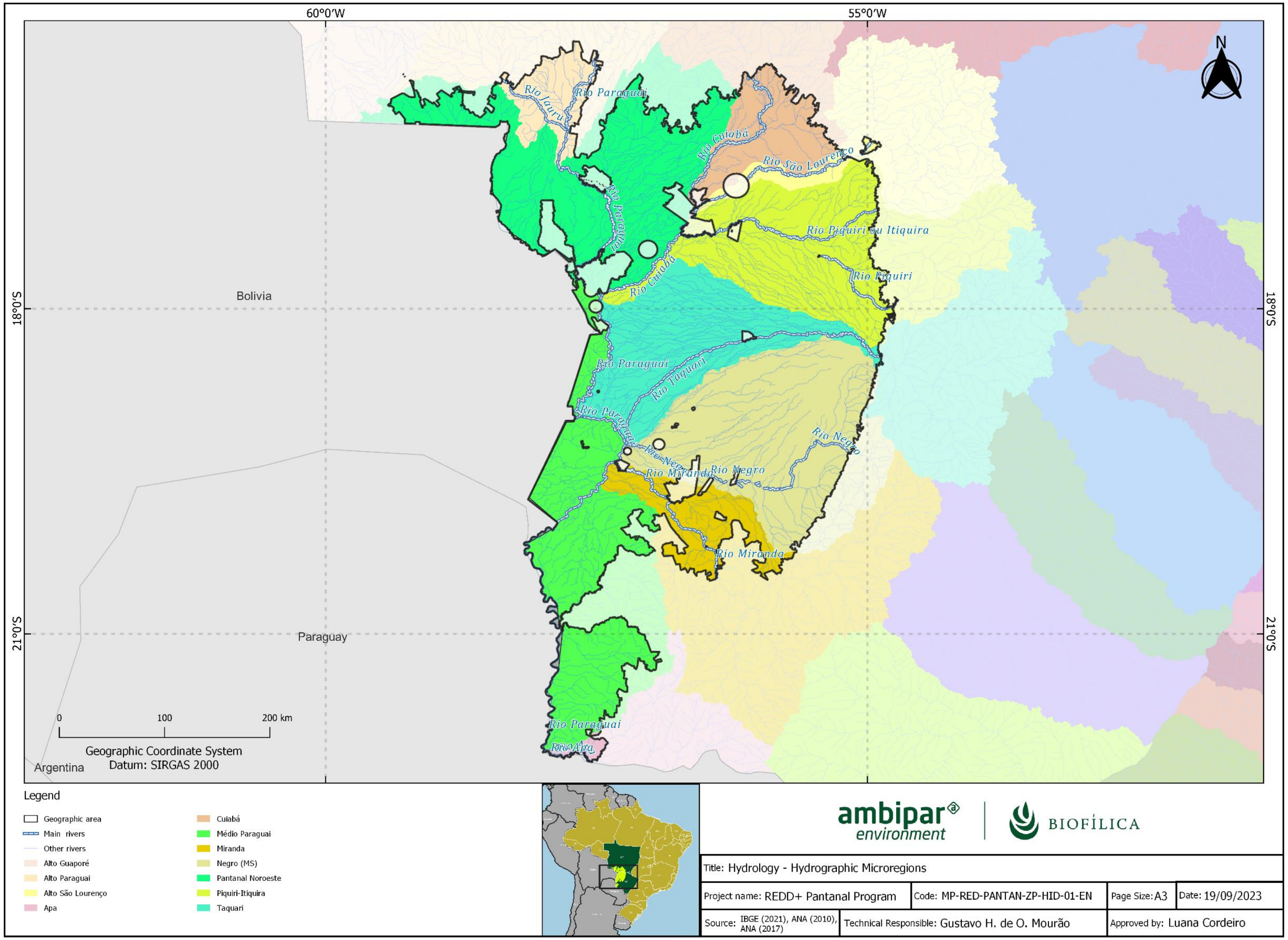

Figure 9. Hydrography and Hydrographic Mesoregions of the Geographic Area of the Pantanal REDD+ Program

Vegetation

The Pantanal biome represents the largest part of the Geographical Area, covering 91.7% of its total extension. Consequently, the predominant phytophysiognomies of the Geographical Area are characterized by the main phytophysiognomies of this biome. These include the following phytophysiognomies: Park Savannah, Wooded Savannah, Grassy-Woody Savannah, Forested Savannah and Savannah-Steppic, which occupy 79% of the geographical area. When added to livestock (pastures), they occupy 94% of the geographical area.

The Savannah, in its various manifestations, is a xeromorphic vegetation adapted to seasonal climates. The Forested Savannah, or Cerradão, is typical of sandy areas with deep soils and features woody vegetation of varying size, often resembling seasonal forests. The Wooded Savannah varies in density and is shaped by annual fires, with a floristic composition similar to the Cerradão, but adapted to different geographical regions. The Park Savannah resembles an "English Park", with predominant grasses and sparse trees, and is found throughout Brazil, with regional variations. The Grassy-Woody Savannah is dominated by grasses and woody plants adapted to fire and grazing. Finally, the Savannah-Steppic encompasses thorny and deciduous grassland vegetation, distributed in various regions of the country, with the Caatinga do Sertão Árido Nordestino being the most representative, characterized by its irregular rainfall (IBGE, 2012). The forest types are shown in Table 6 and on the map in Figure 10.

Table 6. Forest typologies recorded in the Geographical Area of the Pantanal REDD+ Program, based on the classification of Brazilian vegetation (IBGE, 2021d)

Types of vegetation	Area (ha)	Area (%)
Savannah Park	3.585.281	27,62
Wooded Savannah	2.224.156	17,13
Grassy-Woody Savannah	1.987.857	15,31
Forested Savannah	1.586.570	12,22
Grassy-Woody Savannah-Steppic	870.757	6,71
Alluvial Seasonal Semi Deciduous Forest	752.133	5,79
Wooded Savannah-Steppic	405.172	3,12
Savanna-Steppic Park	356.173	2,74
Forested Savannah-Steppic	303.013	2,33
Continental water body	230.592	1,78

Types of vegetation	Area (ha)	Area (%)
Lowland Semi Deciduous Seasonal Forest	217.708	1,68
Lowland Seasonal Deciduous Forest	126.537	0,97
Submontane Seasonal Deciduous Forest	111.543	0,86
Alluvial Evergreen Seasonal Forest	106.718	0,82
Contact (Ecotone)	75.213	0,58
Agriculture	17.103	0,13
Secondary Vegetation	10.930	0,08
Lowland Evergreen Seasonal Forest	7.971	0,06

Note: The vegetation classification table excludes the areas of forestry and hydrography that appear on the map.

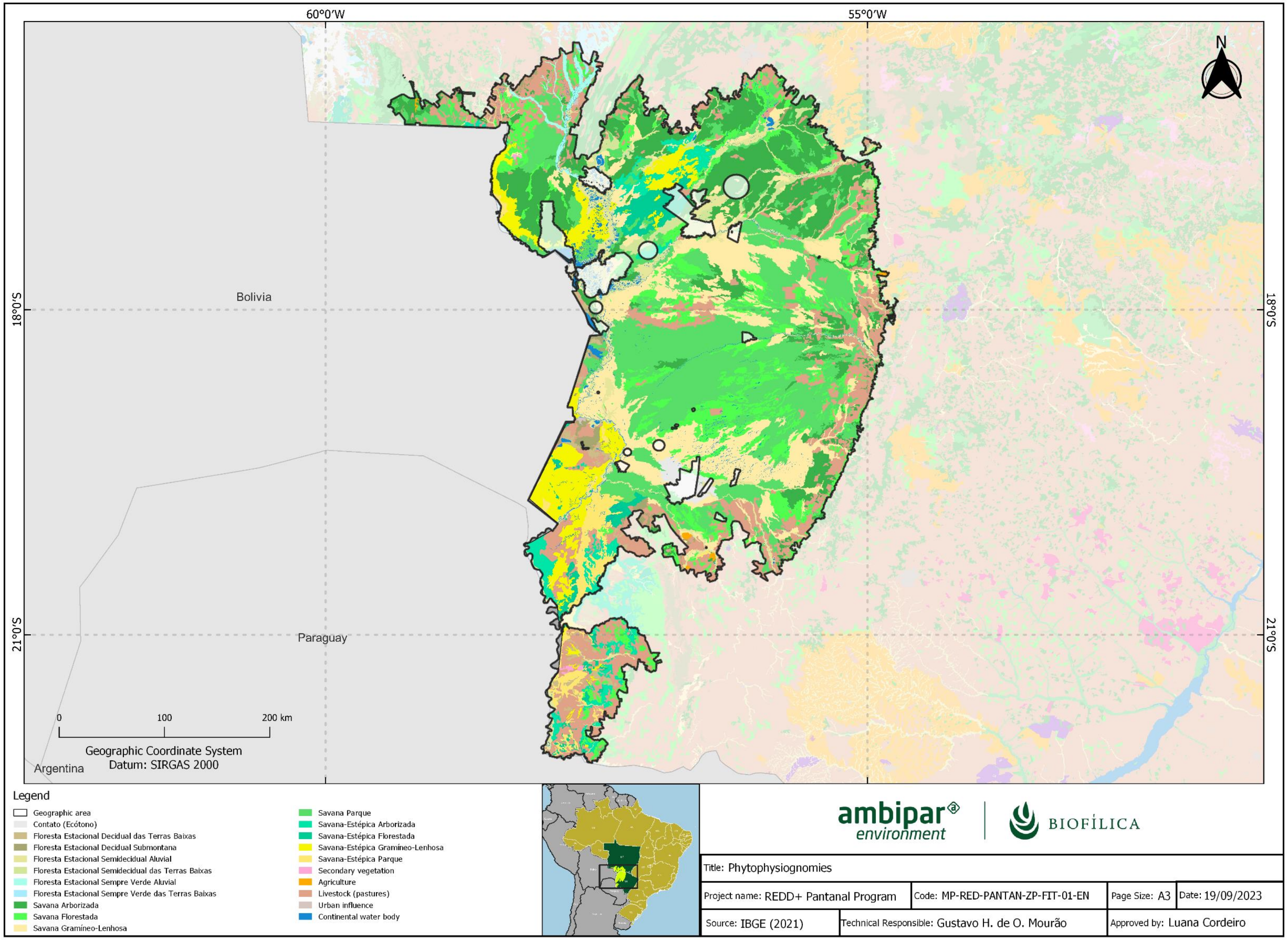

Figure 10. Types of vegetation found in the Geographical Area of the Pantanal REDD+ Program

Climate

The Köppen climate classification system is widely recognized for categorizing global climates based on distinct patterns of temperature and precipitation. Within this system, the main classifications contained in the Geographical Area of the Pantanal REDD+ Program are: Am, Af and Aw, each representing specific subtypes of the tropical climate with unique characteristics (Figure 11).

The rainy tropical climate, denoted as "Af", is characterized by consistent rainfall throughout the year, without a clearly defined dry season. In regions with this climate, each month receives at least 60 mm of rain, and the average monthly temperature always remains above 18°C. On the other hand, the monsoonal tropical climate, classified as "Am", has a short dry season followed by an intense rainy season, usually associated with the monsoon phenomenon. The dry season in this climate lasts a month or less. Precipitation during the driest month is less than 60 mm, but still exceeds the value calculated by the formula (100 - [annual total in mm/25]). The temperature, as in the Af climate, remains consistently above 18°C throughout the year. Finally, the tropical savannah climate, or "Aw", is distinguished by a rainy season and a well-demarcated dry season. Rainfall in the driest month is less than 60 mm and falls below the value determined by the formula (100 - [annual total in mm/25]). The average monthly temperature, similar to the other two classifications, exceeds 18°C during all months (ALVARES, *et al.*; 2014).

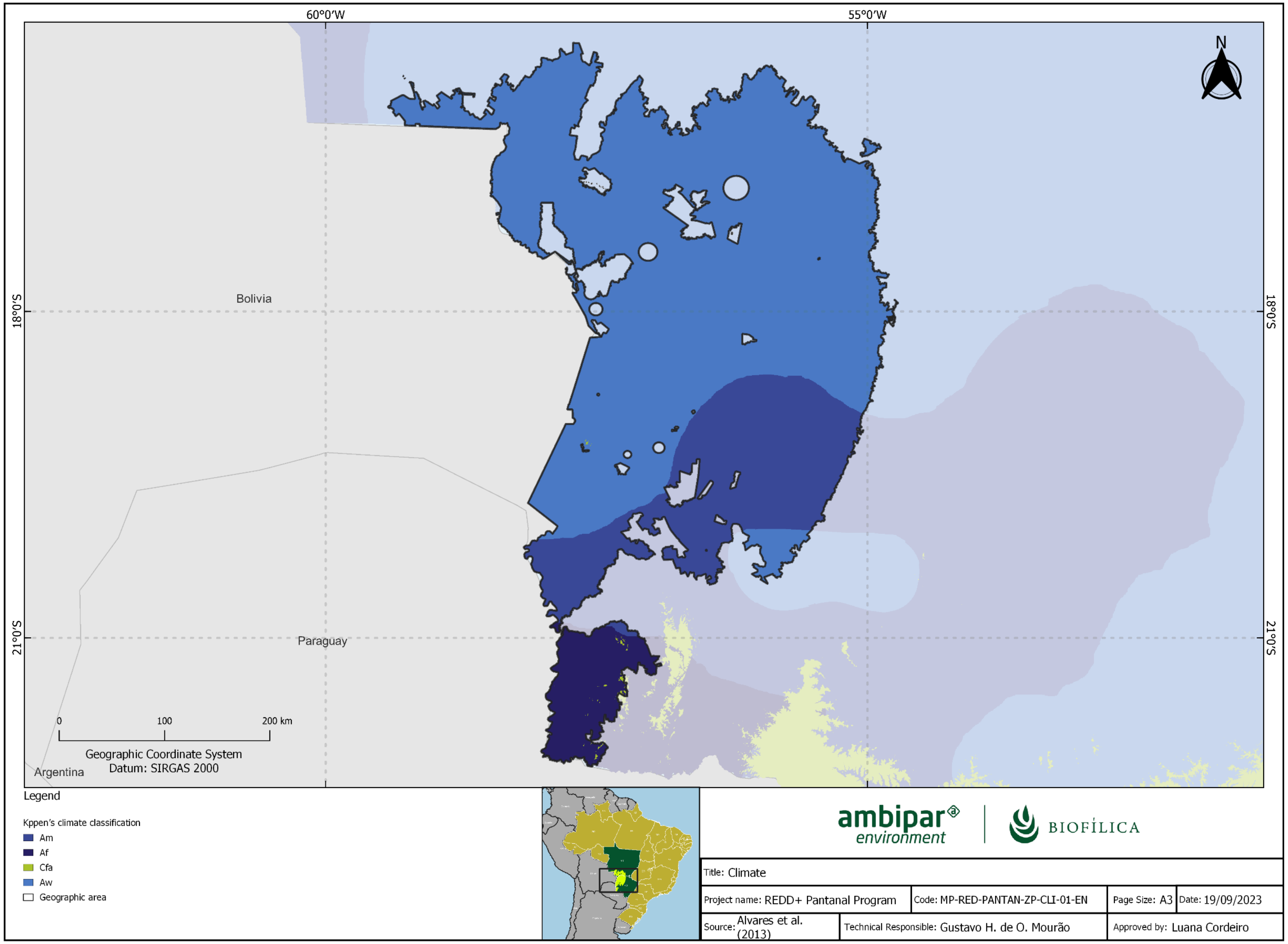

Figure 11. Köppen classification of the climate in the Geographical Area of the Pantanal REDD+ Program

3.4.2 PART 1: CALCULATING ANNUAL AREA OF LAND DEFORESTED

Under development.

3.4.3 PART 2: BASELINE CARBON STOCK CHANGE

Under development.

3.4.4 PART 3: GREENHOUSE GAS EMISSIONS

Under development.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

3.5.2 Additionality Methods

Under development.

3.5.2.1 Step 1 - Identifying alternative land use scenarios

Under development.

3.5.2.2 Step 2 - Investment analysis

Under development.

3.5.2.2 Step 4 - Analysis of common practices

Under development.

3.6 Methodology Deviations

Not applied.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

Under development.

4.1.1 Prospecting new areas for inclusion in the Pantanal REDD+ Program

The prospecting of new areas for the Pantanal REDD+ Program is carried out through three mechanisms, described in more detail below:

1. Partnerships with institutions that bring together various rural landowners, such as trade unions and sectoral associations;
2. Partnerships with prospectors and area brokers; and
3. Word of mouth (buzz marketing).

Partnerships with institutions that bring together several landowners

Biofílica Ambipar has a partnership with institutions with a significant presence in the state of Mato Grosso do Sul that bring together several landowners or rural properties, such as rural unions, sector associations and research bodies, which are interested in promoting the carbon program to their members as an alternative for conservation and sustainable income generation that is compatible with the Pantanal's traditional way of life.

Workshops and lectures to publicize the carbon program are organized with the support of these institutions, with the aim of broadening the reach of the initiative and clearing up doubts among farmers, who are generally almost completely unaware of the issue and, in some cases, have had negative experiences with frivolous carbon proposals or those that didn't materialize.

Campaigns to attract new areas are also carried out on social media relevant to the sector (e.g. WhatsApp).

Partnerships with prospectors and area brokers

Area prospecting service providers and/or real estate brokers are engaged to indicate rural properties located in the Pantanal biome of Mato Grosso do Sul and with native vegetation in excess of the minimum required by current environmental legislation.

Word of mouth Advertising

Option 1: Owners of areas already participating in the carbon program and those studying the opportunity to join are encouraged to publicize the opportunity and bring more areas into the initiative, as they understand that there will be a gain in scale with benefits for all participants. In this way, these owners spread the word, talking to neighbors, relatives and acquaintances, and inspiring in them an interest in evaluating the adherence and potential of their own farms for a possible carbon project.

Option 2: Owners of areas that are already participating in the carbon program or are considering it, spread the word about the opportunity by talking to neighbors, relatives and acquaintances, inspiring their interest in assessing the suitability and potential of their own farms for a possible carbon project.

Opção 2: Proprietários de áreas que já participam do programa de carbono ou que estão analisando a possibilidade de participação divulgam a oportunidade em conversa com vizinhos, parentes e conhecidos, inspirando nestes o interesse em avaliar a aderência e potencial de suas próprias fazendas para um eventual projeto de carbono.

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Under development.

5.2 Project Emissions

Under development.

5.3 Leakage Emissions

Under development.

5.3.1 STEP 1: Determination of the Baseline Rate of Forest Clearance by the Deforestation Agent

Under development.

5.3.2 STEP 2: Estimate New Projection of Forest Clearance by the Baseline Agent Of Deforestation with Project Implementation if No Leakage is Occurring

Under development.

5.3.3 STEP 3: Monitor All Areas Deforested by Baseline Agent of Deforestation Through the Years in Which Planned Deforestation was Forecast To Occur

Under development.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Under development.

Vintage period		Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated buffer pool allocation (tCO_2e)	Estimated total VCU issuance (tCO_2e)
14/10/2021	13/10/2022	168.861	0	0	37.149	131.712
14/10/2022	13/10/2023	173.274	0	0	38.120	135.153
14/10/2023	13/10/2024	177.686	0	0	39.091	138.595
14/10/2024	13/10/2025	182.099	0	0	40.062	142.037
14/10/2025	13/10/2026	186.511	0	0	41.033	145.479
14/10/2026	13/10/2027	190.924	0	0	42.003	148.921
14/10/2027	13/10/2028	195.337	0	0	42.974	152.363
14/10/2028	13/10/2029	199.749	0	0	43.945	155.804
14/10/2029	13/10/2030	204.162	0	0	44.916	159.246
14/10/2030	13/10/2031	208.574	0	0	45.886	162.688
Total		**1.887.777**	**0**	**0**	**415.179**	**1.471.998**

6 MONITORING

6.1 Data and Parameters Available at Validation

Under development.

6.2 Data and Parameters Monitored

Under development.

6.3 Monitoring Plan

Under development.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Under development.

7.2 Baseline Emissions

Under development.

7.3 Project Emissions

Under development.

7.4 Leakage Emissions

Under development.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Under development.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Under development.

Section	Information	Justification

APPENDIX 2: LIST OF LAND DOCUMENTS COLLECTED

NO.	Documents/Item	Laws/Regulations
1	Updated real estate records ("Full Content Registry Certificate") indicating any encumbrances, debts or lawsuits related to the property(ies) in the last 20 (twenty) years.	Law 10.406/2002 - Brazilian Civil Code Law 6.015/1973 - Public Records Law
2	Certificate issued by the competent Real Estate Registry Office with the complete chain of title (CADEIA DOMINIAL) transferring it from Public to Private Domain.	Federal Constitution/1988 - Brazilian Federal Constitution Law 10.406/2002 - Brazilian Civil Code Law 6.015/1973 - Public Records Law Law 601/1850 - Law on Vacant Lands
3	CCIR (Rural Property Register) with INCRA	Law 4.504/1964 - Land Statute
4	Geodetic survey approved by INCRA, if applicable	Law 6.015/1973 - Public Records Law
5	Declaration and proof of payment of the Tax on Rural Territorial Property (ITR) for the last 5 years. ITR Information and Calculation Document - DIAT, for each ITR Declaration made available.	Law 9.393/1996 - ITR Law - Tax on Rural Territorial Property
6	Proof of registration of the property in the Rural Environmental Registry (CAR), with a description of the area of native vegetation, preservation areas and identification of Legal Reserve areas.	Law 12.651/2012 - Brazilian Forest Code
7	Negative Debt Certificate and Environmental Infringement Notice issued by the competent State Environmental Agency. [1] Depending on the jurisdiction, the certificate issued by SEFAZ may cover environmental fines.	Law 9.605/1998 - Federal law on environmental crimes
8	Certificates issued by the Federal and State Public Prosecutor's Offices - Environmental Prosecutors, indicating the (in)existence of legal proceedings involving the Property, activities developed or under development, or the existence of environmental damage subject to compensation or recovery measures (NEGATIVE CERTIFICATE FROM THE STATE AND FEDERAL MINISTRY).	Law 7.347/1985 - Law on Public Civil Action
9	Certificates issued by the State Court of Justice (CERTIDÃO NEGATIVA DOS DISTRIBUIDORES DA JUSTIÇA ESTADUAL), for the location of the Property(ies) and the domicile of the Owner(s), covering a period of 10 (ten) years.	Law 10.406/2002 - Brazilian Civil Code

APPENDIX 3: EVIDENCE OF SDG INDICATORS

Under development.